

GENERAL SCIENCE GRADE 10: LINEAR MOTION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 3.0: Natural Physical Science
Sub-Strand	Sub-Strand 3.2: Linear Motion
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Terms used in linear motion (distance, displacement, speed, velocity, acceleration) – Equations of linear motion with constant acceleration (no derivations needed) – Examples of linear motion with constant acceleration – Freely falling bodies and the effect of gravity
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain terms used in linear motion; b) calculate variables of motion using equations of linear motion with constant acceleration; c) investigate the effect of gravity on bodies under free fall; d) appreciate the significance of linear motion and free fall in real life.
Core Competencies	– Communication and Collaboration: as the learner works with peers to discuss and present in plenary the application of linear motion in real life. – Learning to Learn: as the learner carries out assignments on examples of bodies under free-fall motion and presents the findings in a group discussion.
Core Values	– Peace: the learner displays tolerance and calmness when discussing the application of linear motion in real life. – Unity: the learner displays cooperation and collaboration with others while investigating the effect of gravity on objects.
Science & Engineering Practices	– Asking Questions and Defining Problems: as the learner formulates questions about why dropped objects speed up and how gravity influences free fall near Nairobi and other locations. – Planning and Carrying Out Investigations: as the learner drops objects from measured heights to observe increasing speed and infer the effect of gravity. – Analysing and Interpreting Data: as the learner uses collected timing data from free-fall observations to calculate acceleration due to gravity and compare with $g = 10 \text{ m/s}^2$. – Using Mathematics and Computational Thinking: as the learner solves numerical problems on linear motion using the equations of motion with constant acceleration. – Constructing Explanations: as the learner explains observed differences in the speed of objects dropped from different heights using the concept of gravitational acceleration. – Obtaining, Evaluating, and Communicating Information: as the learner researches applications of linear motion and free fall

	using digital and print media and presents findings to peers.
Pertinent & Contemporary Issues (PCIs)	– Road Safety: increased awareness on how to minimise fatal injuries resulting from road crashes on sloping roads using the concept of free fall of objects (e.g., lorries descending the steep Kinangop escarpment or the Limuru–Nairobi road). – Life Skills Education (Self-Awareness): the learner observes safety measures when dropping objects to investigate the effect of gravity on objects.
Career Connections	– Civil and Structural Engineering: understanding linear motion and free fall is essential for designing safe roads, bridges, and buildings in Kenya's expanding infrastructure (e.g., the Nairobi Expressway, SGR). – Sports Science and Athletics: Kenyan elite runners and jumpers (e.g., athletes from the Rift Valley) rely on principles of velocity, acceleration, and displacement for performance analysis and coaching. – Aviation and Aerospace Technology: free-fall calculations and motion equations underpin pilot training and the growing Aviation track at Senior School. – Traffic and Road Safety Management: NTSA officers and road engineers apply motion equations to set speed limits and design warning systems on steep Kenyan roads. – Medicine and Biomechanics: understanding free fall and impact forces informs injury prevention and rehabilitation for falls — relevant to Kenya's hospitals and physiotherapy sector.
Focus for Lessons	This unit develops learners' ability to describe, quantify, and apply the principles of linear motion — from defining key terms to solving real-world problems using the equations of constant acceleration and investigating free fall. Instruction anchors physics concepts in Kenyan everyday contexts such as road safety on escarpment roads, the flight path of a maize cob dropped from a granary, and the sprint mechanics of Rift Valley athletes. Learners progress from qualitative understanding of motion vocabulary to quantitative problem-solving and experimental investigation of gravitational acceleration.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does a runaway mkokoteni on the Limuru escarpment keep getting faster — and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes? KICD KEY INQUIRY QUESTIONS: 1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Anchoring Phenomenon	A mkokoteni (hand-cart) loaded with sacks of maize is accidentally released from rest at the top of the steep Limuru–Nairobi escarpment road and rolls freely downhill. Witnesses describe how it starts moving slowly but becomes dangerously fast within seconds, eventually crashing. Students view a short video clip of a runaway vehicle on a Kenyan escarpment road and are asked: "Why does the cart keep getting faster even though no one is pushing it, and could we have predicted how fast it would be going before it crashed?" This is puzzling because the cart has no engine, yet its speed continuously increases — suggesting an invisible force (gravity) is continuously doing work on it. The phenomenon directly motivates the need to define motion terms precisely, develop equations that can predict the cart's speed and distance at any moment, and understand free fall as the purest expression of gravitational acceleration.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Motion Vocabulary: Students watch a video clip of a runaway vehicle on a steep Kenyan escarpment road. They record initial observations and predictions about why the vehicle accelerates. The teacher facilitates a whole-class discussion to surface prior knowledge and misconceptions. Students are introduced to and distinguish between distance, displacement, speed, velocity, and acceleration using the mkokoteni scenario and Kenyan athletics examples. A class Driving Question Board (DQB) is created and initial questions are posted. Lesson 2 — Equations of Linear Motion (Part 1): Using the mkokoteni context, the teacher introduces the four equations of linear

	motion with constant acceleration (presented without derivation as per KICD). Students identify the five variables (u, v, a, s, t) and practise selecting the correct equation for given problems. Worked examples use Kenyan road and athletics contexts (e.g., a matatu accelerating from a bus stop on Thika Road). Students solve structured numerical problems in pairs. Lesson 3 — Equations of Linear Motion (Part 2 — Problem Solving): Students work in small groups to solve a set of progressively challenging problems using all four equations of motion, including contexts such as a Rift Valley athlete's 400 m race, a boda-boda decelerating before a police check, and a ball thrown upward from a Mombasa rooftop. Groups present solutions on the board; the teacher uses cold-calling (minimum 3 groups per problem set) and provides 30-second wait time before taking responses. The DQB is updated with new insights. Lesson 4 — Investigating Free Fall (Predict & Observe): Students first predict: "If you drop a ripe mango and a pebble from the same height, which hits the ground first?" They record predictions individually. The class then drops objects (rubber bung, crumpled paper, stone) from a fixed height of 2 m and observes that all reach the ground at the same time (in the absence of significant air resistance). Students infer that gravity accelerates all objects equally. Data on drop height and time are recorded and used to estimate g. Lesson 5 — Freely Falling Bodies (Explain & Calculate): Students use their experimental data from Lesson 4 and the equations of linear motion (with $u = 0$, $a = g \approx 10 \text{ m/s}^2$) to calculate the speed of the mkokoteni at various points along the escarpment, the time taken for the mango to fall from the tree, and the height from which an object was dropped given its impact speed. The teacher explicitly connects each calculation back to the anchoring phenomenon. Students discuss the dangers of free fall on sloping surfaces and steep roads in Kenya. Lesson 6 — Applications, Road Safety & Real-Life Contexts: Using the equations and free-fall principles, students investigate the braking distance of a matatu travelling at different speeds on Mombasa Road. They discuss road safety (NTSA guidelines, signage on steep roads such as the Kinangop descent) and how understanding linear motion can save lives. Students research (using digital devices or print media) one additional real-life application of linear motion in Kenya (e.g., SGR train scheduling, pole-vault athletics, construction safety) and prepare a brief group presentation. Lesson 7 — Group Presentations & Model Building: Groups present their real-life application research from Lesson 6 (3 minutes per group; teacher cold-calls 2 members per group for follow-up questions). The class collaboratively builds and refines a visual explanatory model (annotated diagram or motion-time graph sequence) connecting the anchoring phenomenon — the mkokoteni on the escarpment — to all sub-strand concepts: motion terms, equations, free fall, gravity, and road safety mitigation. The DQB is reviewed and students identify which questions have been answered and which remain. Lesson 8 — Consolidation, DQB Review & Assessment: Students individually solve a structured set of problems revisiting all four equations and free fall, set in the context of the anchoring phenomenon (the mkokoteni) and supporting phenomena (Kenyan sprinter, falling mango). The class conducts a final DQB review — remaining questions are discussed and answered collectively. Students reflect in their journals on how their thinking about the anchoring phenomenon has changed since Lesson 1, articulating the role of gravity, constant acceleration, and the equations of motion in explaining and predicting the mkokoteni's dangerous descent.
Supporting Phenomena	– A ripe mango falling from a tree on a Kisumu farm — students notice it falls faster and faster rather than at a constant speed, raising the question of what determines how quickly it accelerates. – A Kenyan 100 m sprinter (e.g., a school athletics champion) filmed from the starting blocks — the video shows the athlete accelerating from rest to top speed, illustrating real-world non-uniform linear motion with measurable displacement and time

	intervals. – A water tank filling on a Nairobi rooftop where a pipe drips at constant intervals — drops photographed with a phone stroboscope show increasing spacing between successive positions, providing visual evidence of acceleration under gravity. – A matatu braking to a stop on Mombasa Road — passengers lurch forward, prompting discussion of deceleration (negative acceleration), braking distance, and road safety.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 x 40-minute lessons)): The Runaway Mkokoteni: Why Does It Keep Getting Faster?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon of the runaway mkokoteni, to give learners precise definitions of distance, displacement, speed, velocity and acceleration, and to establish the claim that the cart's velocity keeps increasing (it accelerates), while opening the Driving Question Board.
Knowledge	By the end of the lesson, the learner should be able to explain terms used in linear motion (distance, displacement, speed, velocity and acceleration): distance is total path length (scalar, m); displacement is straight-line change in position with direction (vector, m); speed is distance/time (scalar); velocity is displacement/time (vector); acceleration is change in velocity per unit time (m/s^2). Learners also state that the cart's speed rises by 2 m/s every second, so it accelerates.
Skills	Observing and recording data from a video; making and justifying predictions; asking scientific questions; distinguishing scalar from vector quantities using a 400 m lap; developing an initial model (annotated sketch) of the cart's motion; communicating ideas in whole-class discussion.
Attitudes	Appreciate the significance of linear motion in real life and road safety; respect others' opinions during discussion (Respect); show tolerance and calmness when ideas differ (Peace); cooperate with peers in groups (Unity). Core competencies: Communication and collaboration; Learning to learn. PCI: Road safety.
Key Inquiry Question	How is free fall motion important in real life? How can the dangers of free fall in our day-to-day life be mitigated? (Lesson driving question: Why does a runaway mkokoteni on the Limuru escarpment keep getting faster, and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?)
Purpose in Storyline	This is the launch lesson. It introduces the phenomenon of Mzee Kamau's runaway mkokoteni (300 kg loaded, released from rest on the Limuru-Nairobi escarpment, crashing into a parked lorry after 100 m in 10 s). It surfaces prior ideas, gives the vocabulary needed for all later lessons, and creates the DQB. Lesson 2 will build on the definitions of velocity and acceleration to develop equations that let learners predict the cart's speed and distance.
Safety Notes	Use a 400 m track or marked school field only in a controlled way for the walking and running demonstration; keep the learners walking, not racing. Keep the area clear of vehicles and obstacles. In the discussion, treat the road crash sensitively: nobody was hurt in the incident, only the cart was destroyed, and some learners may have experienced road accidents. Do not let learners test anything on real slopes or with real carts.

B. LESSON OVERVIEW

Learners meet the phenomenon: Mzee Kamau, a vegetable and maize seller, was helping a customer when his loaded mkokoteni (cart 50 kg + 5 sacks of maize at 50 kg each = 300 kg) rolled away from rest down a steep straight section of the Limuru-Nairobi escarpment road. The bystander Achieng says, 'It started slowly then flew.' It crashed into a parked lorry after 100 m, about 10 s later. Learners predict why it speeds up with no engine and no one pushing, watch a short video of a runaway vehicle on a Kenyan escarpment road, and record observations. They then use the cart and Kenyan athletics (a 400 m lap) to distinguish distance from displacement and speed from velocity, and are introduced to acceleration as the rate at which velocity changes. The class uses the average speed of 10 m/s (100 m in 10 s) and the idea that the cart's speed rises by 2 m/s every second, without any equations yet. The lesson ends with the class building the Driving Question Board and a first annotated sketch model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Position vs. time graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Average velocity and average speed from graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners individually read the story of Mzee Kamau's mkokoteni and Achieng's remark, 'It started slowly then flew.' In their science notebooks they sketch a graph of speed against time that they think matches the cart (no formulas needed), and write 2 to 3 sentences answering: 'Why does the cart keep getting faster even though no one is pushing it, and could we have predicted how fast it would be going before it crashed?' They also write one number guess: how fast (km/h) was it going when it hit the lorry? They share with a partner for 2 minutes and then record a class poll of guesses on the board.	Read the story aloud slowly. Say: 'Mzee Kamau turns away for a moment. The cart has no engine. Nobody is pushing. What do you think happens next?' Give individual silent WAIT TIME of 10-15 seconds after the main question before anyone speaks, then 4 minutes of writing. Circulate and photograph or note 3 contrasting predictions (for example 'it speeds up because of gravity', 'the maize weight pushes it', 'it keeps one steady speed'). Cold-call 4 learners, including at least 2 who have not volunteered, with: 'Tell us your prediction and one reason.' Do not say which is correct. Record each prediction on the board under the heading 'Our first ideas'. Say: 'Hold on to these. We will check them against evidence.'	Predict-then-test with a claim and reason (Asking Questions and Defining Problems; Developing and Using Models): committing to a written prediction and a sketch graph before instruction makes prior ideas visible, so that learners can later see how their thinking changed, and gives the teacher data on misconceptions such as 'a moving object needs a continuing push' or 'heavier objects speed up more'.	Look for: (1) each learner has written a prediction with a reason; (2) the sketched graph is either flat (constant speed), rising, or something else, which diagnoses whether the learner expects speeding up; (3) the number guess in km/h shows their sense of scale. Tally on a quick checklist how many predicted 'speeds up', 'steady speed', or 'speeds up because heavy'. Note any learner who refers to gravity, and any who says the cart needs a push to keep moving. Do not grade; use only to plan the discussion.
Observe Phase  VIDEO: Position vs. time graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Average velocity and average speed from graphs Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners watch a short video clip (about 2 minutes, played twice) of a runaway vehicle on a steep Kenyan escarpment road (for example a clip shared by the teacher from a news report; if no video is available, the teacher uses a prepared picture strip of the cart at 0, 2, 4, 6, 8 and 10 s positions, with the spacing between pictures increasing). In groups of four they complete an observation table with three columns: 'What I saw', 'What I noticed about the speed at the start, middle and end', and 'What I wonder'. They then record, as a class, the facts of the case: starting from rest, slope steep and straight, no engine, no one	Before playing say: 'Watch the first two seconds and the last two seconds. Do not write during the first viewing; just watch the speed.' Play once with no pause, then say: 'Turn to your group and say in one sentence what you noticed about the speed.' After the second viewing give 3 minutes to fill in the table. Circulate with the question: 'How do you know it is going faster, and what are you comparing?' Press with: 'Is it covering more distance each second, or the same?' Give WAIT TIME of 10-15 seconds after asking: 'What is the evidence that speed is changing?' Cold-call 3 groups to share one observation each, and write them on the board	Structured observation with data comparison (Analyzing and Interpreting Data): learners separate what they saw from what they infer, and using picture spacing and average speed they find the pattern that speed keeps increasing. The pattern becomes the phenomenon-level claim that the cart is accelerating.	Check the observation tables for: (1) at least one observation stating that speed changes (for example 'slow at first, fast at the end'); (2) separation of observation from explanation; (3) correct use of the facts that average speed is $100 \text{ m} / 10 \text{ s} = 10 \text{ m/s}$. Listen for groups that notice growing gaps between positions. Ask exit-ticket style thumbs: 'Thumbs up if you think the cart went at the same speed throughout.' Count the hands. Reteach with the picture strip if more than a quarter agree.

	pushing, crash after 100 m in about 10 s. The teacher shows the average speed: 100 m in 10 s gives 10 m/s on average, and reveals that the cart was moving at about 72 km/h at the end, so it must have been much slower at the start. Learners compare the picture spacing to see that the distance covered in each second keeps growing.	under 'Evidence'. Reveal the facts: 100 m, about 10 s, average speed 10 m/s. Ask: 'If the average is 10 m/s and it ended at 20 m/s, what does that tell us about the start?' Wait 10-15 seconds. Let learners reason that it started at 0 and rose steadily. Say: 'Its speed went up by about 2 m/s every single second.' Do not give the equations; say only that Lesson 2 will show how to predict this.		
Explain Phase  VIDEO: Deriving displacement as a function of time, acceleration, and initial velocity <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Instantaneous velocity and instantaneous speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners build the precise motion vocabulary in three steps. (1) The 400 m lap: the class goes outside (or imagines the track) with an athlete in the Rift Valley style 400 m race; a volunteer walks one full lap of a marked 400 m (or scaled) track. Learners decide the distance (400 m, total path) and displacement (0 m, because they end where they started) and discuss why the numbers differ. (2) Speed and velocity: learners compare 'speed is distance/time' with 'velocity is displacement/time, with direction', then use the cart: 'The cart moves 100 m down the slope in 10 s: its average speed is 10 m/s, and its velocity is 10 m/s down the slope.' They convert 72 km/h to 20 m/s by dividing by 3.6. (3) Acceleration: they use the statement that the cart's velocity rises by 2 m/s every second to define acceleration as change in velocity per unit time, in m/s^2, so the cart's acceleration is 2 m/s^2. They fill a Frayer-style vocabulary card for each of the five terms with definition, scalar or vector, unit, a cart example and a Kenyan athletics example. In pairs they classify a list of statements (for example 'Mzee Kamau's cart travelled 100 m', 'the cart moved	Say: 'A Kenyan 400 m runner finishes a full lap. How far did she run, and how far is she from where she started?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners (one from each side of the room). Write 'distance = 400 m' and 'displacement = 0 m' side by side and ask: 'Why can both be true?' Then ask: 'If she took 50 s, what is her average speed? And her average velocity?' Wait 10-15 seconds; guide to 8 m/s and 0 m/s. Draw the cart on the slope and say: 'Same question for the cart: it moved 100 m in a straight line down the slope. Is the distance equal to the displacement here? Why?' Then say: 'At the start the velocity is 0. After 1 s it is 2 m/s. After 2 s it is 4 m/s. What do you notice about how much it changes each second?' Wait 10-15 seconds; cold-call 2 learners. Introduce acceleration only as 'change in velocity each second', and write '$a = 2 \text{ m/s}^2$' and what it means in words. Do not derive or use the equations of linear motion; say: 'Next lesson we will learn how to use this to predict the cart's speed at any moment.' Address misconceptions: 'Speed and acceleration are not the same: a fast cart can have zero	Contrasting cases and Frayer vocabulary cards (Using Mathematics and Computational Thinking; Constructing Explanations): placing distance and displacement, and speed and velocity, side by side with the 400 m lap lets learners see exactly what changes between the two, and connecting each definition to the cart anchors the abstract terms to the phenomenon.	Look for: (1) learners correctly state the 400 m lap as distance 400 m and displacement 0 m; (2) the vocabulary cards give correct units (m, m/s, m/s^2) and mark distance and speed as scalar, displacement, velocity and acceleration as vector; (3) in the classification task, at least 80% of statements are placed correctly; (4) the final claim mentions that velocity increases steadily. Use a quick whiteboard check: 'Show me the unit of acceleration.' Note learners who write m/s instead of m/s^2 and revisit with 'change in velocity (m/s) every second'.

	100 m down the slope', 'the cart's speed was 10 m/s on average', 'the cart's speed rose by 2 m/s each second') into the correct term. Finally, they write a one-sentence claim: 'The cart accelerates because its velocity increases steadily from zero.'	acceleration, and a cart can have a small speed with large acceleration.' Use one question to probe: 'At the very first moment the velocity is zero. Is the acceleration also zero? Explain.' Cold-call 2 learners. Emphasise respect: 'Listen to each idea, then build on it.'		
Driving Question Board (DQB) Creation  VIDEO: Introduction to reference frames <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Displacement During Constant Acceleration (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	The class builds the Driving Question Board on a wall chart headed with the driving question: 'Why does a runaway mkokoteni on the Limuru escarpment keep getting faster, and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?' Each learner writes 2 questions on sticky notes or strips about the phenomenon (for example 'Why does gravity make it faster?', 'Would a heavier cart go faster?', 'How fast would it be after 5 s?', 'What if it fell straight down?', 'How can drivers on steep roads like Kinangop stay safe?'). In groups of four they sort questions into categories (for example 'Why does it speed up?', 'How can we predict?', 'Free fall and gravity', 'Road safety'). Groups post their questions on the DQB and each group reads out one favourite. The two KICD inquiry questions are added: 'How is free fall motion important in real life?' and 'How can the dangers of free fall in our day-to-day life be mitigated?'	Say: 'You now know the facts of the case and some new words. What are you still wondering? Write questions that start with Why, How, What if.' Give 3 minutes of silent writing, then WAIT TIME of 10-15 seconds after asking: 'Which questions can we only answer by measuring or calculating?' Model a good question: 'How could we calculate the speed after 5 s?' Show how to rewrite a weak question ('Is gravity strong?') into a stronger one ('How does gravity change the speed of the cart each second?'). Cold-call 4 groups to read one question each and ask the class: 'Which category does this belong to?' Point to the questions that need the equations of linear motion and tell learners these will be answered in Lessons 2 and 3. Link road safety questions to the PCI: 'The NTSA is concerned about runaway vehicles on steep roads, so your road safety questions are real questions.' Keep the board visible and say: 'We will return to this board in every lesson and move questions to the Answered column.'	Driving Question Board with question sorting (Asking Questions and Defining Problems): learners generate and categorise their own questions, which gives purpose to the unit, makes the class's curiosity visible, and lets them see that mathematics is needed to answer 'how fast' and 'how far'.	Look for: (1) every learner posts at least two questions; (2) at least some questions are testable or calculable, not only yes/no; (3) categories chosen by groups are sensible and refer to the phenomenon; (4) learners can say which of their questions need measurement or calculation. Use a quick tally on the DQB categories to see which ideas (gravity, prediction, free fall, safety) the class cares about. Provide a sentence starter ('How could we find out...') to learners who struggle.
Model Building Phase  VIDEO: Motion problems:	Individually, learners draw an initial model of the runaway mkokoteni on a half page of the notebook: an annotated diagram of the slope with the cart at five positions (0 s,	Say: 'A model is a drawing that explains and helps predict. Show the motion, the direction, and what is changing.' Display an example with only the slope and arrows as	Initial model construction and revision (Developing and Using Models): learners combine the diagram, gravity arrow, spacing and speed sketch into one	Check for: (1) spacing between positions increases; (2) a gravity or downhill arrow is drawn along the slope; (3) the caption uses at least three of the terms distance,

finding the maximum acceleration <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Displacement During Uniform Acceleration Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	2 s, 4 s, 6 s, 10 s at increasing spacing), labels for the direction of motion, an arrow showing gravity pulling it down the slope, and a rough speed-time sketch that rises from zero. Using colour, they annotate where the velocity is small and where it is large and write a caption using at least three of the five terms: 'The cart accelerates at 2 m/s^2 because its velocity increases by 2 m/s every second.' In pairs they compare, add one improvement each, and write one thing the model cannot yet do (for example 'it cannot tell us the speed at 7 s '). The sketches are kept in a Model folder to be revised in later lessons. Finish with an exit ticket: 'Write one thing you used to think about the cart that you now think differently, and one question you still have.'	a starting point (not a complete answer). Circulate and ask: 'Which part of your drawing shows that the cart is accelerating?' and 'Where does the gravity arrow go, and why along the slope?' Give WAIT TIME of 10-15 seconds after each question. Cold-call 3 learners to explain their models in 30 seconds; ask the class to give 'one star and one wish'. Highlight that the gaps between cart positions must increase. Close by saying: 'Our model is a first draft. In Lesson 2 we will learn the equations of linear motion so it can predict the cart's speed and position at any time.' Collect exit tickets. Check misconceptions: if a learner draws constant spacing, ask: 'What would that mean for the speed?' Ask learners to keep their predictions from the Predict Phase for comparison in Lesson 8.	representation of the phenomenon, which they can test and improve as new ideas arrive. Naming what the model cannot yet do shows the need for mathematics.	displacement, speed, velocity and acceleration correctly; (4) the speed-time sketch starts at zero and rises; (5) the exit ticket shows a change in thinking. Mark each indicator as Yes/Partly/Not yet on a class list. Learners who draw equal spacing or a flat graph get a brief one-to-one conversation and the picture strip from the Observe Phase.
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D. TEACHER REFLECTION

After the lesson ask yourself: (1) Did learners separate observation from explanation in the video task? (2) Which misconceptions appeared in the predictions (for example the cart needs a continuous push, heavier objects speed up more, speed and acceleration are the same), and how many learners held them? (3) Could learners explain why the 400 m lap has distance 400 m but displacement 0 m? (4) Did I give enough WAIT TIME of 10-15 seconds and cold-call a range of learners, including quieter ones and learners from different groups? (5) Which DQB questions need Lessons 2-3, which need the free fall investigation, and which concern road safety? (6) What changes will I make to pacing, for example less time on the vocabulary cards and more on the model? Record the three most common prediction types to compare with learners' final explanations in Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We watched or studied a runaway mkokoteni (300 kg loaded with maize) released from rest on the Limuru-Nairobi escarpment road. It started slowly, then went much faster, and crashed into a parked lorry after 100 m in about 10 s. The gaps between its positions grew each second, and its average speed was 10 m/s .
What did I learn?	Distance is total path length (scalar, m); displacement is the straight-line change in position with direction (vector, m). Speed is distance/time (scalar); velocity is displacement/time (vector). Acceleration is change in velocity per unit time (m/s^2). A 400 m lap has distance 400 m but displacement 0 m. The cart's velocity rises by 2 m/s every second, so it accelerates at 2 m/s^2 .
How does this explain the	The cart has no engine and no one pushing it, yet it keeps getting faster because gravity pulls it down the slope all the time, so its

phenomenon?	velocity increases steadily from zero. It starts slowly because its velocity begins at zero. We still need mathematics (the equations of linear motion) to predict exactly how fast and how far it travels, which we will build in the next lessons.
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LESSON 2 (80 minutes (2 x 40-minute lessons)): Equations of Linear Motion: Choosing the Right Equation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the four equations of linear motion to turn the cart's constant acceleration idea from Lesson 1 into numbers, so they can predict its speed and distance at any moment.
Knowledge	By the end of the lesson, the learner should be able to: (a) identify the five variables of linear motion (u , v , a , s , t) with their SI units; (b) state the four equations of linear motion for constant acceleration, given without derivation: (1) $v = u + at$; (2) $s = ut + \frac{1}{2}at^2$; (3) $v^2 = u^2 + 2as$; (4) $s = \frac{1}{2}(u+v)t$; (c) select the equation that links the three known variables and the one unknown; (d) check the equations against the first rows of the mkokoteni table ($t = 1$ to 5 s).
Skills	Analysing and interpreting data (NGSS SEP 4); using mathematics and computational thinking (SEP 5): listing knowns and unknown, selecting an equation, substituting with units, and checking that answers are reasonable; constructing explanations (SEP 6).
Attitudes	Perseverance and precision with units; respect for peers' reasoning (Respect); cooperation in pair work (Unity); willingness to check own answers before accepting them. Core competencies: Critical thinking and problem solving; Communication and collaboration. PCI: Road safety.
Key Inquiry Question	KICD Key Inquiry Question 1: How is free fall motion important in real life? (Lesson 2 builds the mathematical tools needed to answer it.) Driving question: Why does a runaway mkokoteni on the Limuru escarpment keep getting faster, and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?
Purpose in Storyline	Lesson 1 gave learners the vocabulary (distance, displacement, speed, velocity, acceleration) and the claim that the cart's velocity keeps increasing. Lesson 2 supplies the prediction tools: the four equations and a method for choosing among them. Learners test the tools on a matatu on Thika Road, then check them against the first rows of the mkokoteni table ($a = 2 \text{ m/s}^2$, $t = 1$ to 5 s). Lesson 3 will extend the method to multi-step problems, deceleration and upward motion. This lesson does not teach free fall, stopping distances or graphs.
Safety Notes	Mostly a desk-based lesson. Keep bags and chairs out of walkways during pair work. If any learner walks a measured 10 m to estimate speed, use the corridor or field only with the teacher present. Calculators are optional; set shared-use rules.

B. LESSON OVERVIEW

Learners start from the question 'Could we have predicted how fast the cart was going before it crashed?' and predict how speed and distance would grow each second. They then meet the matatu on Thika Road ($u = 0$, $a = 1.5 \text{ m/s}^2$, $t = 8 \text{ s}$), where the teacher models the method: list the five variables, mark knowns and the unknown, and pick the equation that contains exactly those four. The four equations are given without derivation, as the KICD design requires. Learners then use them to check the first rows of the mkokoteni table ($t = 1$ to 5 s) against $a = 2 \text{ m/s}^2$, which shows that v increases by 2 m/s each second while s grows as t^2 . In pairs they solve structured problems, justify their equation choice in words, and add questions to the Driving Question Board. The lesson ends with the first update to the class model: the cart's acceleration is constant, so equations can predict its motion.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Choosing kinematic equations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Displacement During Constant Acceleration (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Learners recall the phenomenon: Mzee Kamau's loaded mkokoteni (300 kg) released from rest at the top of the Limuru-Nairobi escarpment, rolling 100 m to a parked lorry. Achieng said 'it started slowly then flew'. Individually, in their notebooks, learners predict: (a) the cart's speed after 1 s, 2 s and 5 s, and (b) the distance covered after 5 s compared with after 10 s. Then they write one sentence: 'To predict this exactly, we would need to know ____.' In pairs they compare predictions and note where they disagree. Learners also recall from Lesson 1 that acceleration is the change in velocity per unit time, and that the cart's velocity keeps rising.	Display the phenomenon image and say: 'Achieng said it started slowly then flew. Today we ask: could Mzee Kamau's neighbours have predicted exactly how fast it would be going?' Ask learners to write predictions silently. WAIT TIME: 12 seconds of silence before pair talk. Say: 'Do not look at your neighbour yet. Write numbers, even if you guess.' Cold-call 4 pairs to share their prediction for speed after 5 s, and record all four on the board without judging. Ask: 'What would you need to know about the cart to be sure?' WAIT TIME: 10 seconds. Cold-call 2 more learners. Record the variables they name (starting speed, time, how quickly speed changes, distance) in a list headed 'What we need to know'. Do not name the symbols yet.	Predict-Share-Compare, with a 'What we need to know' list. Writing numerical predictions first exposes learners' existing ideas (some expect speed to be steady, others expect it to double). The learner-generated list of needed quantities prepares them to see the five variables as meaningful rather than arbitrary symbols.	Observe whether learners write numbers with units (m/s, m) rather than words only. Tally how many predict speed rising by equal amounts each second (consistent with constant acceleration) versus doubling each second or staying steady. Listen for: 'speed keeps going up each second' (linked to acceleration from Lesson 1). Note learners who confuse speed with distance. Check the board list: did learners name starting speed, time and rate of change of speed?
Observe Phase  VIDEO: Average velocity for constant acceleration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Setting up problems with constant acceleration Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(25 min) Part A - Naming the variables: Learners are given a one-page card for the matatu on Thika Road: 'A matatu leaves a stage on Thika Road from rest, accelerates steadily at 1.5 m/s² for 8 s.' The class builds a variable table: u (initial velocity, m/s), v (final velocity, m/s), a (acceleration, m/s²), s (displacement, m), t (time, s). Learners mark each variable as 'known' or 'unknown' for the matatu: u = 0, a = 1.5 m/s², t = 8 s known; v and s unknown. Part B - The four equations: The teacher presents the four equations of linear motion, written exactly: (1) $v = u + at$; (2) $s = ut + \frac{1}{2}at^2$; (3) $v^2 = u^2 + 2as$; (4) $s = \frac{1}{2}(u+v)t$. A note states they apply when acceleration is constant and are given without derivation. Learners underline which variable is	Hand out the Thika Road card. Say: 'Before choosing any equation, we list. Which of the five variables do we know? Which one are we hunting?' Cold-call 3 learners to name one variable each with its unit. WAIT TIME: 10 seconds after each question. Present the four equations on a prepared chart and say: 'These work only when acceleration is constant. We use them without deriving them, as the syllabus asks.' Ask: 'Which variable is missing from equation 3? Why might that help us?' WAIT TIME: 10 seconds, then cold-call 2 learners. Model the matatu on the board while thinking aloud: 'I know u, a, t. I want v. Equation 1 has exactly these four. So I use it.' Write units at every step. Then ask: 'Is 12 m/s reasonable for a matatu after 8 s? Convert: 12 x 3.6	Worked example with think-aloud, then 'List-Choose-Substitute-Check'. Modelling the thinking, not just the algebra, shows how an expert chooses an equation by matching knowns and unknown, which is the lesson's central skill. Testing the equations on the cart table lets learners see that the equations reproduce a pattern they could describe only in words in Lesson 1: velocity rising by 2 m/s each second.	Check that learners (a) list knowns and unknown BEFORE choosing an equation; (b) write units at each step; (c) state the missing-variable test for the equation they chose. Look for common errors: using v instead of u in equation 2; forgetting to square t; dividing by 3.6 in the wrong direction; using a = 10 for the cart. Observable success: pairs correctly complete five table rows (t = 1 to 5 s) with v = 2, 4, 6, 8, 10 m/s and s = 1, 4, 9, 16, 25 m. Collect the Thika Road calculation as a quick exit check for 5 randomly selected learners.

	MISSING from each equation: (1) no s; (2) no v; (3) no t; (4) no a. Part C - Using the method: The teacher models the matatu: to find v, list $u = 0$, $a = 1.5$, $t = 8$, choose equation (1): $v = 0 + 1.5 \times 8 = 12$ m/s (43.2 km/h). To find s, choose equation (2): $s = 0 + \frac{1}{2} \times 1.5 \times 8^2 = 48$ m. Check with equation (4): $s = \frac{1}{2}(0+12) \times 8 = 48$ m. Part D - Test on the cart: Pairs complete the first rows of the mkokoteni table using $u = 0$, $a = 2$ m/s²: $t = 1$ s: $v = 2$ m/s, $s = 1$ m; $t = 2$ s: $v = 4$, $s = 4$; $t = 3$ s: $v = 6$, $s = 9$; $t = 4$ s: $v = 8$, $s = 16$; $t = 5$ s: $v = 10$, $s = 25$. They check that v rises by 2 m/s each second and that s grows as t^2 (1, 4, 9, 16, 25).	= 43.2 km/h.' Circulate during Part D. Say: 'Fill in v first with equation 1, then s with equation 2, and tell your partner which equation you used each time.' Pause the class at $t = 3$ s and ask 3 cold-called learners: 'What do you notice about the v column? And the s column?' WAIT TIME: 12 seconds. Do NOT introduce $t = 6, 8$ or 10 s rows yet; learners may extend them as a challenge. Remind: 'The cart's acceleration is 2 m/s², not 10. The slope only provides a fraction of g. We will explore why in a later lesson.'		
Explain Phase  VIDEO: Deriving displacement as a function of time, acceleration, and initial velocity Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Displacement During Constant Acceleration (Flexbook) Source: CK-12  Search ARES for similar readings	(25 min) Part A - Structured pair problems (Kenyan contexts): Pairs solve three problems, each requiring the learner to write a four-line solution (Knowns, Unknown, Equation chosen and why, Answer with units). Problem 1: A matatu on Thika Road accelerates from rest at 1.5 m/s² for 8 s; find the final velocity (12 m/s) and the displacement (48 m). Problem 2: Mzee Kamau's cart, from rest at $a = 2$ m/s², has covered $s = 25$ m; find the velocity at that point using equation (3) and no time: $v^2 = 2 \times 2 \times 25 = 100$, so $v = 10$ m/s. Learners confirm that this equals the table value at $t = 5$ s. Problem 3: A learner on a Kenyan school field jogs from rest and reaches 4 m/s in 2 s; find the acceleration (2 m/s²) and the distance covered (4 m), choosing from equations (1) and (4) or (2). Part B - Explain in words: Each pair writes 2 to 3 sentences answering: 'Why did the equation you chose work better than the	Pair learners by mixed confidence. Say: 'Partner A writes the knowns and unknown. Partner B chooses the equation and explains why. Then swap for the next problem.' Circulate with a checklist. Ask pairs: 'Which variable is missing from the equation you chose? How do you know that is the right one?' For pairs finishing early, pose: 'Can you solve Problem 1 using a different equation to check your answer?' After 12 minutes, stop the class: 'Pencils down. Turn to your partner and compare answers.' Cold-call 4 pairs to give the answer to Problem 1 and 3 pairs for Problem 2. WAIT TIME: 10 to 15 seconds before each cold call. For a wrong answer say: 'Interesting. Show us your knowns list.' and let the class locate the error rather than telling them. Facilitate a whole-class discussion: 'Why does the distance in the cart table grow so much faster than the speed?' WAIT TIME: 15 seconds, then cold-call 3	Claim-Evidence-Reasoning in pairs: the claim is the numerical answer, the evidence is the list of knowns and the substituted equation, and the reasoning is the written justification for the equation choice. Partner roles (writer and chooser, then swap) ensure that both learners practise the selection decision. Linking back to Achieng's statement turns numerical skill into an explanation of the phenomenon.	Observable indicators: (a) 'Knowns and unknown' list present before any equation in at least 80 percent of written solutions; (b) correct equation chosen in Problems 1 to 3 (Problem 2 should use equation 3 because t is not given); (c) correct final values with units: 12 m/s, 48 m, 10 m/s, 2 m/s², 4 m; (d) written justification that names the missing variable. Use a 3-level tally: independent, with prompting, needs reteach. Plan a short reteach in Lesson 3 for learners who mix up u and v, or ignore the units.

	others?' and 'Which variable did your chosen equation not need?' Part C - Link to the cart: Learners use the table to explain why the cart 'felt sudden': at $t = 5$ s, $v = 10$ m/s and $s = 25$ m, but the distance covered doubles twice over by $t = 10$ s ($s = 100$ m), because s grows as t^2. They write one sentence connecting this to Achieng's 'started slowly then flew'.	learners. Close by asking: 'What did the equations let us do that Lesson 1's words could not?' Elicit: predict exact values.		
Driving Question Board (DQB) Creation  VIDEO: Choosing kinematic equations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Setting up problems with constant acceleration Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(10 min) The DQB from Lesson 1 is displayed. Learners first review the questions already posted. Individually they write on sticky notes: (a) one question that Lesson 2 has ANSWERED or partly answered (for example, 'Can we predict the cart's speed at 5 s?' now answered: 10 m/s) and (b) one NEW question that the lesson raised (for example, 'What if the cart did not start from rest?', 'What if the matatu slows down instead?', 'Why is the cart's acceleration only 2 m/s² and not 10?', 'How are these equations used when an object falls straight down?'). They place notes on the DQB under the headings 'Answered/Partly answered' and 'New questions'. The class groups new questions by theme: slowing down, falling objects, the slope.	Say: 'Look at the board from Lesson 1. Which questions did today's equations help with?' WAIT TIME: 12 seconds. Cold-call 3 learners to move a note to 'Answered' and say why. Say: 'Now write a question that today made you curious.' Collect notes and cluster them aloud. Highlight themes that connect to later lessons without teaching them: 'Questions about slowing down and objects thrown upward will come in Lesson 3. Questions about falling will come in Lessons 4 and 5. Why only 2 m/s² for the cart is a question we will return to.' Cold-call 2 learners to read a new question aloud.	Driving Question Board: sorting questions into 'answered' and 'new' makes learning visible and shows learners that science progresses by answering questions and raising better ones. Clustering questions by theme shows how each upcoming lesson connects to the driving question.	Look at whether notes distinguish what was answered (for example 'speed at 5 s is 10 m/s') from what remains open. Quality of new questions: do they extend the equations to new conditions (deceleration, falling, non-zero start)? Note learners who write only vague questions ('What is acceleration?') and prompt them to refer to numbers. Check that at least 5 questions connect to free fall or the effect of the slope.
Model Building Phase  VIDEO: Average velocity for constant acceleration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	(10 min) Learners begin the class model of the cart in their notebooks. Model v1 (from Lesson 1): a labelled diagram of the cart on the slope with the note 'velocity keeps increasing'. Learners now add: (a) the five variables labelled for the cart: $u = 0$ m/s, $a = 2$ m/s², and the table values for $t = 1$ to 5 s; (b) a short box titled 'Prediction tool' that lists the four equations of linear motion, and a one-line rule	Say: 'Open your Lesson 1 model. We are not starting again. We are adding a prediction tool.' Model the first addition on the board: write $u = 0$ and $a = 2$ m/s² on the cart diagram. Ask learners to complete the rest independently. WAIT TIME: 10 seconds, then circulate. Cold-call 3 learners to read their one-sentence statement about constant acceleration and prediction. Ask: 'Which part of	Cumulative model revision (annotated diagram plus a prediction tool). Adding quantitative tools to the qualitative Lesson 1 model shows how mathematical reasoning strengthens an explanation. The 'Still to explain' box makes gaps visible, which motivates Lessons 3 to 5.	Check that each model includes: (a) $u = 0$ and $a = 2$ m/s² labelled on the diagram; (b) at least two correct table values (for example $v = 10$ m/s and $s = 25$ m at $t = 5$ s); (c) the four equations written correctly; (d) a sentence linking constant acceleration to predictability. Watch for the misconception that $a = 10$ m/s² for the cart, or that the equations apply to any motion. Use a 3-point

 HTML: Constant Acceleration and Inertia (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	'List knowns and unknown, then choose the equation that has those four variables'; (c) one sentence: 'Because acceleration is constant, we can predict v and s at any moment.' They also write in a 'Still to explain' box: 'Why is a only 2 m/s^2?' and 'What if the cart slowed down or went up?' Learners complete the summary table (observed, learned, explained) for homework.	today's evidence supports your sentence?' Elicit: 'The table: velocity rose by 2 m/s every second, and s equalled t^2.' Remind learners that the equations apply when acceleration is constant, and that the cart on the slope has $a = 2 \text{ m/s}^2$, not 10. Assign homework: complete the summary table and solve one extra problem: a matatu on Thika Road from rest at $a = 1.5 \text{ m/s}^2$ travelling $s = 48 \text{ m}$, find v using equation (3) (answer 12 m/s).		check (labelled diagram, equations, sentence) as a quick tally.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners list knowns and unknown BEFORE reaching for an equation, or did they hunt for numbers to plug in? (2) Which pairs could justify their choice using the missing-variable idea, and which just copied? (3) How many learners wrote $a = 10$ for the cart or mixed up u and v ? (4) Were the table values for $t = 1$ to 5 s ($v = 2$ to 10 m/s , $s = 1$ to 25 m) enough for learners to see v rising by 2 m/s each second and s growing as t^2 ? (5) Did the DQB notes show curiosity about slowing down and falling? Use these answers to plan a brief reteach at the start of Lesson 3, which builds on this lesson's selection method with the sign convention, negative acceleration and upward motion.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe or work out today? (Examples: the matatu on Thika Road reaches $v = 12 \text{ m/s}$ and $s = 48 \text{ m}$ after 8 s ; the mkokoteni table for $t = 1$ to 5 s gives $v = 2, 4, 6, 8, 10 \text{ m/s}$ and $s = 1, 4, 9, 16, 25 \text{ m}$.)
What did I learn?	What did we learn about the equations of linear motion? (Include the five variables u, v, a, s, t with units, the four equations, and the rule for choosing: list the knowns and the unknown, then pick the equation that has exactly those four variables.)
How does this explain the phenomenon?	How does this help explain the driving question? (Use the cart's constant acceleration of 2 m/s^2 to explain why the velocity rises by 2 m/s every second, why the cart felt sudden, and how the equations let us predict its speed and distance at any moment.)

LESSON 3 (80 minutes (double lesson of 2 x 40 minutes)): Problem Solving with the Equations of Linear Motion: Slowing Down and Going Upward

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the four equations of linear motion in multi-step problems. They apply a sign convention, handle negative acceleration (slowing down) and treat an object thrown upward under $a = -10 \text{ m/s}^2$. This moves them toward predicting the runaway mkokoteni's motion and understanding free fall.
Knowledge	By the end of the lesson, the learner should be able to: calculate variables of motion using equations of linear motion with constant acceleration ($v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$; $s = \frac{1}{2}(u+v)t$), including cases where acceleration is negative and where motion is vertical and upward ($a = -10 \text{ m/s}^2$).
Skills	Using Mathematics and Computational Thinking: listing knowns and the unknown, choosing an equation, and substituting with units. Analysing and Interpreting Data: checking answers against a second equation and against common sense. Constructing Explanations: defending a solution to the class. Communication and collaboration: presenting and critiquing solutions in groups.
Attitudes	Appreciate the significance of linear motion in real life (the runner, the boda-boda, the thrown ball). Show Peace by staying calm and tolerant when peers challenge a solution. Show Unity by sharing roles in the group (reader, calculator, checker, presenter).
Key Inquiry Question	How is free fall motion important in real life? How can the dangers of free fall in our day-to-day life be mitigated? (Lesson focus: how do we correctly handle slowing down and upward motion so that the same equations that predict the runaway mkokoteni also predict a braking boda-boda and a ball thrown upward?)
Purpose in Storyline	Lesson 2 gave learners the four equations and a method for choosing one. Lesson 3 stress-tests that method on harder cases: starting from rest, braking to a stop, and rising then falling. Learners see that one set of equations, with a clear sign convention, covers every constant-acceleration case. This prepares them to investigate free fall (Lesson 4) and to predict the mkokoteni at any moment, which is the driving question.
Safety Notes	The Mombasa rooftop ball is a paper problem. Do not throw objects from roofs or windows. If you demonstrate an upward throw, use a soft ball outdoors in a cleared area, with learners standing well back and no one underneath. Keep walkways clear during group work and gallery walks. Remind learners that a real boda-boda must never be braked hard on a wet or loose-surface road without care, because that is a lesson in road safety, not an experiment.

B. LESSON OVERVIEW

Learners return to the runaway mkokoteni on the Limuru escarpment. Mzee Kamau's loaded cart (300 kg) is released from rest and gains 2 m/s every second ($a = 2.0 \text{ m/s}^2$). Learners now ask what happens when acceleration is not a simple speed-up from rest. They predict answers to three Kenyan situations, then solve them in groups with a strategy: draw a sketch, choose the positive direction, list knowns and the unknown, choose an equation, solve, and check with a second equation. The problems are (a) a Rift Valley 400 m runner accelerating from rest, (b) boda-boda rider Otieno braking before a police check, and (c) a ball thrown upward from a 20 m Mombasa rooftop. Groups present on the board with cold-calling and a 30-second wait time. Learners then explain why the sign of acceleration matters, add new questions to the DQB, and add a sign-convention note to the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: linear motion equations sign  HTML: Solving Real-World Problems Using Multi-Step Equations (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners recall the runaway mkokoteni: released from rest, $a = 2.0 \text{ m/s}^2$, 10 m/s after 5 s , 20 m/s (72 km/h) after 10 s and 100 m . They are shown three situations on the board: (a) a Rift Valley runner leaving the blocks in a 400 m race, (b) boda-boda rider Otieno braking from 15 m/s when he sees a police check, (c) a ball thrown upward at 15 m/s from a Mombasa rooftop 20 m above the ground. Individually, they write predictions in their journals: (1) Will the runner's speed after 2.5 s be less than, equal to, or more than 10 m/s ? (2) Will Otieno's boda-boda stop in less than 20 m , about 40 m , or more than 60 m ? (3) How long will the ball stay in the air, and will it hit the ground faster or slower than 15 m/s ? Each learner gives a reason and a confidence rating from 1 to 5. They then share with a partner.	Say: "Remember Mzee Kamau's mkokoteni. It sped up by 2 metres per second every second. Today we meet situations where things slow down, or go up before they come down. Before any calculation, write what you think will happen." Read each scenario aloud slowly. After each question say: "Take 15 seconds of silent thinking time. Pens down until I say write." WAIT TIME: 15 seconds, then 1 minute to write. Do not confirm answers. Cold-call 4 learners with different predictions, using the name cards, and ask each: "What in the situation made you choose that?" Record the predictions in a tally on the board. Say: "We will come back to these predictions at the end."	Predict-Observe-Explain (Predict step) with a confidence rating. Committing to a prediction and a reason exposes preconceptions, such as 'negative acceleration means going backwards' or 'a thrown ball has zero acceleration at the top'. These are the ideas the later calculations will confirm or challenge.	Check journals for a written prediction and reason for all three situations. Note how many learners (tally) believe the ball has no acceleration at its highest point, and how many believe slowing down means a negative speed. Look for learners who link braking to the mkokoteni in reverse. Learners who leave a reason blank get a prompt: "What do you know about how the cart's speed changed?"
Observe Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: linear motion equations sign  EXERCISE: Solving Real-World Problems Using Multi-Step Equations Practice	Part 1 (10 min): The teacher models a 6-step strategy with the mkokoteni as a warm-up: (1) sketch, (2) choose positive direction (down the slope is positive, so $a = +2 \text{ m/s}^2$), (3) list u , v , a , s , t with signs and units, (4) mark the unknown, (5) choose the equation that contains the unknown and three knowns, (6) check with a second equation. Using $t = 5 \text{ s}$, learners confirm $v = 10 \text{ m/s}$ and $s = 25 \text{ m}$. Part 2 (30 min): In groups of 4 with roles (reader, recorder, calculator, checker), learners solve the Problem Set. (a) Runner: from rest, $a = 4 \text{ m/s}^2$ for 2.5 s . Find v	Model the strategy on the board with the mkokoteni, thinking aloud: "Down the slope is my positive direction, so a is plus 2." Then circulate with a checklist. Ask groups that are stuck: "What are your five variables? Which one do you not know? Which equation has the unknown and no other missing variable?" For (b) ask: "What is v when the boda-boda stops? Is a positive or negative, and why?" For (c) ask: "What is the velocity at the very top? What is the sign of a on the way up, at the top, and on the way down?" Give a hint card only after two attempts. Do not give final answers. WAIT TIME:	Structured Problem-Solving Protocol (Using Mathematics and Computational Thinking), with a Sign-Convention Sketch. A fixed routine forces learners to decide the positive direction first. This makes negative acceleration meaningful (the velocity decreases) rather than mysterious. Checking with a second equation builds the habit of validating a model's prediction, as will be needed for the mkokoteni.	Observe whether each group (i) draws a sketch with an arrow marking positive, (ii) writes $a = -3 \text{ m/s}^2$ and $a = -10 \text{ m/s}^2$ with the negative sign, (iii) writes $v = 0$ for stopping and at the top of the throw, and (iv) checks with a second equation. Expected values: (a) $v = 10 \text{ m/s}$, $s = 12.5 \text{ m}$; (b) $t = 5 \text{ s}$, $s = 37.5 \text{ m}$; (c) 1.5 s up, 11.25 m extra, 31.25 m maximum height, reaches the ground at $t = 4 \text{ s}$, impact velocity -25 m/s . Tick a checklist per group. Note common errors: using $+15$ for a , treating -25 m/s as slower than 15 m/s , or forgetting that the rooftop displacement is -20 m , not $+20 \text{ m}$.

<i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	and s. (b) Boda-boda: $u = 15 \text{ m/s}$, $a = -3 \text{ m/s}^2$, finds the time to stop ($v = 0$) and the stopping distance. (c) Ball: thrown upward at $u = +15 \text{ m/s}$ from a 20 m rooftop, up is positive, $a = -10 \text{ m/s}^2$. Find (i) the time to the highest point, (ii) the extra height risen, (iii) the maximum height above the ground, (iv) the time to reach the ground ($s = -20 \text{ m}$), and (v) the impact velocity. Each group fills a solution sheet showing the sketch, knowns, equation, working and a check. Early finishers extend: compare with the mkokoteni by asking which of the three situations has acceleration in the same direction as the motion.	after each question to a group, wait 10 seconds before adding a hint. Pause at 20 minutes and say: "Checkers, verify with a second equation, for example $v^2 = u^2 + 2as$." Keep time and remind: "Ball problem: you may not throw anything; this is on paper only."		
Explain Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: linear motion equations sign  HTML: Solving Real-World Problems Using Multi-Step Equations (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Each group puts its solution to problems (a), (b) and (c) on a poster or the board. The class does a gallery walk, then whole-class presentations. For each problem, at least 3 groups are cold-called to present, ask questions and challenge each other. Learners then explain in their own words: (1) why is a negative for the boda-boda and for the ball, but positive for the mkokoteni? (2) Why does the ball take 1.5 s to rise but 4 s to reach the ground rather than 3 s? (3) What does the answer -25 m/s mean, and is it faster or slower than the initial 15 m/s? (4) How does Otieno's stopping distance connect to the dangers on a sloping road like the Limuru escarpment? Learners write a 3-sentence explanation using the words 'sign', 'direction' and 'acceleration', then compare their final answers with their Predict journals and mark each prediction as supported or revised.	Say: "Before any group speaks, everyone check your own working for 30 seconds." WAIT TIME: give 30 seconds of silent thinking before taking responses on each problem. Cold-call a minimum of 3 groups per problem using a random-name method: "Group 4, what answer did you get for the boda-boda stopping distance? Group 7, do you agree? Group 2, what equation did you use to check?" Compare any differing answers on the board and ask the class to find the error: "Where does the working differ?" Press for reasoning: "How do you know a is negative?" and "Why does the ball have a $= -10$ at the very top, where its velocity is zero?" Contrast with the mkokoteni: "On the slope a $= +2$ because the cart speeds up in the positive direction. Otieno slows in the positive direction, so a is negative." Pose the safety link: "If Otieno was travelling at 15 m/s but the police check was 30 m away, what happens?" (37.5 m needed,	Argumentation from Evidence (Engaging in Argument from Evidence) with a Cold-Call Share-Out. Groups defend a numerical answer with working. Comparing different answers for the same problem produces productive disagreement that exposes misconceptions about signs. Linking the three problems back to the mkokoteni shows that the equations and the sign convention are general, supporting transfer to new motion situations.	Listen for learners who can say that negative acceleration means velocity is decreasing in the chosen positive direction, not that the object is moving backwards. Listen for a justification that the ball's total time is 4 s because it rises 1.5 s, then falls 31.25 m (and not 20 m) to the ground. Look for learners who can state that impact velocity is -25 m/s, meaning 25 m/s downward. Use a quick 3-question exit slip (individual): (1) Write the value of a for a ball thrown upward. (2) Write the value of v at the highest point. (3) In 'a = -3 m/s²', what does the minus sign tell us? Sort slips into 'secure', 'nearly', 'not yet' piles for follow-up.

		so he would not stop in time). Ask 3 learners to restate the sign rule in their own words. Avoid evaluating until at least 3 groups have spoken; thank groups for correcting errors.		
Driving Question Board (DQB) Creation  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: linear motion equations sign  EXERCISE: Solving Real-World Problems Using Multi-Step Equations Practice Related topic: not an exact match for this lesson. Source: CK-12  Search ARES for similar readings	Learners revisit the class DQB, started in Lesson 1 around the question: 'Why does a runaway mkokoteni on the Limuru escarpment keep getting faster, and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?' In pairs they: (1) put a tick on existing questions that today's work helps answer (for example, how can equations predict speed and distance?), (2) write new questions on sticky notes, such as: Why is the mkokoteni's a only 2 m/s^2 while the ball's a is 10 m/s^2 ? If the ball is dropped from the roof instead of thrown, how long would it take? How far does a matatu travel when it brakes at speed? Could a driver stop a runaway cart by braking? Why does the ball hit the ground faster than it was thrown up? (3) Place new notes beside the categories 'equations', 'slowing down', 'going upward/free fall' and 'safety'. They also add one fact to the 'What we know' column.	Say: "Look at our board. Which questions did today's problems help us answer? Which new questions do you have?" Give pairs 3 minutes with sticky notes. WAIT TIME: 10 seconds of silent thinking before pairs write. Cold-call 3 pairs to read out a new question and say where to place it, asking: "What evidence from today made you ask this?" Use the 'free fall' notes to preview Lesson 4: "Several of you ask whether all objects fall at the same rate. We will test that next lesson." Group questions into themes, and explicitly mark questions answered by the equations and those that need investigation. Do not answer the new questions yet.	Driving Question Board with Question Sorting (Asking Questions and Defining Problems). Sorting questions into themes shows learners how today's evidence connects to the driving question, tracks what is answered and not yet answered, and sets the direction of the remaining lessons (free fall, stopping distance, graphs).	Check that each pair has at least one new question linked to today's work, and that questions go beyond recall (for example, 'why' or 'what if' questions). Note whether learners connect the negative-acceleration cases back to the mkokoteni. Success looks like a sticky note reading something like 'If the cart decelerated like the boda-boda, how far would it travel?' Tally question categories to decide what Lesson 4 should emphasise.
Model Building Phase  VIDEO: No closely matching video in the ARES library for this activity.  Search ARES for videos Search terms: linear motion	Learners revise the class model of the mkokoteni (annotated diagram plus velocity-time graph begun in earlier lessons). Individually, then in groups, they add: (1) a sign-convention note: 'down the slope is positive, so $a = +2.0 \text{ m/s}^2$ '; (2) a small annotated panel comparing the three situations: runner ($a = +4 \text{ m/s}^2$), boda-boda ($a = -3 \text{ m/s}^2$), and the thrown ball ($a = -10 \text{ m/s}^2$)	Say: "Our model must explain more than the cart. Add what you learned today about direction and sign." Hand out the class model sheet and ask groups to add the sign-convention panel. WAIT TIME: 10 to 15 seconds of quiet thinking before writing the 'rule of thumb'. Cold-call 3 learners to read their rule of thumb, and ask the class to critique: "Does this rule	Developing and Using Models (Cumulative Model Revision). Adding the sign-convention panel to the existing mkokoteni model turns the model into a general tool for all constant-acceleration motion. Comparing 'I used to think / Now I think' statements makes the revision of ideas explicit.	Check that each model includes (i) the positive direction marked, (ii) $a = +2.0 \text{ m/s}^2$ for the cart, (iii) $a = -3 \text{ m/s}^2$ for the boda-boda and $a = -10 \text{ m/s}^2$ for the ball, and (iv) a sensible rule of thumb (a and v with opposite signs means slowing down). Collect the 'I used to think / Now I think' statements and look for at least one revised idea per learner.

equations sign  HTML: Solving Real-World Problems Using Multi-Step Equations (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	on the way up and down); (3) a note in the mkokoteni diagram that speeds up in the positive direction means a and v share the same sign, while slowing down means they have opposite signs. They write one 'rule of thumb' in their own words and one line of the 6-step strategy they will use in future problems. Finally they revisit their Predict journals and write: 'I used to think ... Now I think ...'.	work for the runner? For the ball at the top?" Point to the mkokoteni: "If someone had braked the cart hard, which sign would a have?" Close by saying: "Next lesson we investigate whether all objects fall with the same acceleration. Keep your ball-problem answers; we will compare."		
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Which sign-convention errors were most common (using +15 for a , -25 m/s seen as 'slower', forgetting $s = -20$ m for the rooftop)? (2) Did at least 3 groups present each problem, and did the 30-second wait time produce more reasoned answers? (3) Did learners link the boda-boda's stopping distance (37.5 m) to the dangers on a sloping road? (4) Which learners still treated the ball's top-of-flight acceleration as zero, and what additional task would help? (5) Did the group roles keep every learner working, or did the calculator dominate? (6) What should be adjusted before Lesson 4 (free fall investigation), where the sign convention for downward motion will be reused?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We solved the Problem Set. The Rift Valley runner reached 10 m/s after 2.5 s and covered 12.5 m. The boda-boda travelling at 15 m/s with $a = -3$ m/s ² stopped after 5 s and 37.5 m. The ball thrown upward at 15 m/s from the 20 m Mombasa rooftop rose 11.25 m more in 1.5 s (31.25 m above the ground), reached the ground at $t = 4$ s and hit it at -25 m/s (25 m/s downward).
What did I learn?	We learned to pick a positive direction first, list the knowns with signs, and choose the equation that has the unknown. Slowing down gives a negative a (opposite sign to v). For a ball thrown upward, $a = -10$ m/s ² all the time, even at the top, where $v = 0$. The same equations work for speeding up, slowing down and vertical motion, just like for Mzee Kamau's mkokoteni ($a = +2.0$ m/s ² down the slope).
How does this explain the phenomenon?	The mkokoteni keeps getting faster because its acceleration is in the same direction as its motion ($a = +2.0$ m/s ² down the slope), so its velocity grows by 2 m/s every second. A braking boda-boda has a negative acceleration because it opposes the motion, so its velocity falls. Because acceleration is constant in each case, the equations of linear motion let us predict speed, distance and time at any moment, and a consistent sign convention is what makes these predictions correct.

LESSON 4 (80 minutes (2 x 40-minute lessons)): Investigating Free Fall: Predict and Observe

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To gather first-hand evidence that all objects fall with the same acceleration when air resistance is negligible, and to estimate g from measured drop height and fall time. Free fall is the extreme case of the mkokoteni's situation: gravity pulling on a vertical path instead of a gentle slope.
Knowledge	By the end of the lesson the learner should be able to: (a) investigate the effect of gravity on bodies under free fall by dropping objects from a fixed height of 2.0 m; (b) state that when air resistance is negligible all objects fall with the same acceleration; (c) use $g = 2h/t^2$ (from $s = \frac{1}{2}at^2$ with $u = 0$) to estimate g from class data ($h = 2.0$ m, mean $t = 0.64$ s, $g = 9.8$ m/s ²); (d) explain why $g = 10$ m/s ² is used in calculations.
Skills	Planning and carrying out investigations (NGSS SEP 3): controlling variables, repeating trials and measuring time with a stopwatch. Analysing and interpreting data (SEP 4): calculating means and g . Using mathematics (SEP 5): rearranging $s = \frac{1}{2}at^2$ to $g = 2h/t^2$. Constructing explanations (SEP 6): inferring that gravity accelerates all objects equally. Evaluating error: reaction-time uncertainty.
Attitudes	Core competencies: Communication and collaboration (learners share roles and report data); Learning to learn (learners compare predictions with evidence and revise ideas). Values: Unity (cooperation while investigating the effect of gravity on objects); Peace (tolerance when predictions differ). PCIs: Life Skills Education (self-awareness: observing safety measures when dropping objects); Road safety (linking gravity on slopes to crashes).
Key Inquiry Question	KICD Key Inquiry Question 1: How is free fall motion important in real life? (Introduced here through evidence that gravity accelerates every object equally. Key Inquiry Question 2 on mitigating the dangers of free fall is addressed in later lessons.)
Purpose in Storyline	Lesson 1 launched the mkokoteni puzzle and the idea that its velocity keeps increasing (it accelerates). Lesson 2 gave the four equations of linear motion, including $s = \frac{1}{2}at^2$ with $u = 0$. Lesson 3 gave the sign convention and $a = -10$ m/s ² for upward motion. In Lesson 4 learners collect their own evidence about gravity: do all objects speed up equally, and what is the value of g ? This sets up Lesson 5, where g is applied to freely falling bodies and compared with the cart's slope acceleration of 2.0 m/s ² .
Safety Notes	Drop from 2.0 m only over a cleared floor or soft sand tray. Use a rubber bung, stone and crumpled paper ball; keep the stone small (about the size of a mango seed). The dropper stands on a stable bench or the teacher stands on a stable chair or steps, with a spotter. Everyone else stands at least 1 m back. No one places hands or feet under a falling object. The stone is placed gently back by the timer after it comes to rest. Wear closed shoes. Spills or rolling objects are retrieved only after the trial is complete.

B. LESSON OVERVIEW

The lesson returns to Mzee Kamau's runaway mkokoteni on the Limuru-Nairobi escarpment road: starting from rest, it reached 20 m/s (72 km/h) after 10 s and 100 m. Learners recall that gravity pulls it down the slope all the time. They then ask what happens when the slope is vertical, which is free fall. They first predict individually which of a ripe mango and a pebble hits the ground first and record the reasoning. In groups they drop a rubber bung, a stone and a crumpled paper ball from 2.0 m, time 5 trials each, and find the mean times: bung 0.63 s, stone 0.64 s, paper ball 0.65 s (mean 0.64 s). The times are almost identical, so the objects accelerate equally despite very different masses. Learners use $s = \frac{1}{2}at^2$ with $u = 0$, rearranged to $g = 2h/t^2$, to estimate $g = 4/0.64^2 = 9.8$ m/s². They discuss reaction-time error and why calculations use $g = 10$ m/s² (theory gives $t = 0.63$ s). Learners add questions to the Driving Question Board and add free fall evidence to their draft model. The mkokoteni is not in free fall because the slope gives it only part of g ; this is explored fully in Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 02 Freely Falling Objects And Acceleration Due To Gravity Source: Seavuria  Search ARES for similar videos  HTML: Acceleration Due to Gravity (Flexbook) Source: CK-12  Search ARES for similar readings	(About 15 minutes) Learners recall the anchoring phenomenon: Mzee Kamau's loaded mkokoteni (300 kg) rolled down the Limuru-Nairobi escarpment and reached 20 m/s (72 km/h) after 10 s and 100 m. Achieng said, 'It started slowly then flew.' The teacher shows a ripe mango and a pebble held at shoulder height and asks: 'If the escarpment were vertical, like a cliff at Hell's Gate, and you dropped a ripe mango and a pebble from the same height, which hits the ground first?' Individually, in their science journals, learners write (1) their prediction (mango first, pebble first, or together), (2) a reason in one sentence, and (3) a quick sketch showing which object speeds up more. Then learners compare predictions with a partner and tally the class votes by raised hands. Common ideas are expected: heavier objects fall faster because gravity pulls harder on them. Learners also predict whether a crumpled paper ball and a flat sheet of paper will fall at the same rate, and why.	Say: 'Remember Mzee Kamau's cart. Nothing pushed it, yet it kept getting faster. Today we ask what happens when the slope is as steep as it can possibly be.' Hold up the mango and the pebble. Do NOT drop them yet. Say: 'Write your prediction silently first. No talking for 2 minutes.' Circulate and note the reasons that learners give. Then ask for a show of hands for each option and record the vote on the board. Cold-call 3 learners with different predictions: 'Tell us your reason, not just your answer.' Give WAIT TIME of 10-15 seconds after each question before calling on anyone. Say: 'Is it true that a heavier object has a bigger pull from gravity? Does a bigger pull mean a bigger speeding up? Hold that thought.' Do not correct any ideas. Say: 'Keep your predictions. We will test them with measurements.'	Predict-Observe-Explain (POE) with individual commitment: writing a prediction and reason before seeing evidence makes learners' current ideas visible, so that later conflict between prediction and data produces conceptual change. Think-Pair-Share lets learners rehearse and justify reasoning safely before public voting.	Collect journals and check each prediction for: (a) a clear choice, (b) a reason that mentions mass, size or air, and (c) a sketch. Indicators: learners who write that heavier objects fall faster hold the misconception to be revisited; learners who mention air or shape show useful ideas. Tally the vote on the board: expect a majority to choose the pebble or the heavier object. Note names of learners who predict 'together' to invite their reasoning later.
Observe Phase  VIDEO: 02 Freely Falling Objects And Acceleration Due To Gravity Source: Seavuria  Search ARES for similar videos  EXERCISE:	(About 30 minutes) In groups of 5 with assigned roles (dropper, timer, spotter, recorder, reporter), learners test their predictions. Equipment: a metre rule or tape and string to mark a fixed drop height of $h = 2.0$ m (measured from the bottom of the object to the floor), a stopwatch or phone timer, a rubber bung, a small stone and a crumpled paper ball. They also drop a flat sheet of paper beside a	Demonstrate safe dropping first: 'Stand behind the line. The dropper releases, does not throw. Spotter, your job is to watch the floor area.' Say: 'Fair test: what must we keep the same?' Cold-call 2 learners (height, release method). Give WAIT TIME of 10-15 seconds. Show how to release both objects at once from the same height and ask learners to listen for the landing sounds: 'Do	Data table and pattern-finding with controlled variables: holding the height constant and varying only the object isolates the effect of mass and shape, and repeated trials with averaging teach how to handle measurement uncertainty. Error discussion helps learners distinguish a real difference from reaction-time noise.	Observe whether each group (a) measures a 2.0 m height accurately, (b) releases objects without pushing, (c) records 5 trials for each object, and (d) calculates means correctly. Indicators: learners say that 0.01 to 0.02 s differences are within reaction-time error; learners keep the height the same for all trials. Ask each recorder to show the table: a complete table with units (s) shows

Acceleration Due to Gravity Practice Source: CK-12  Search ARES for similar readings	crumpled one as a fair test of shape. For each of the three objects they drop 5 times from 2.0 m, time each fall, and record in a table: object, trial 1 to 5, mean time. The class data are collected on the board: rubber bung mean 0.63 s, stone 0.64 s, crumpled paper ball 0.65 s; mean of all three 0.64 s. Learners notice that the times differ by only 0.01 to 0.02 s even though the stone is much heavier than the paper. They discuss what makes timing difficult (reaction time of about 0.1 to 0.2 s when starting and stopping the watch) and why repeating trials and averaging helps. The flat sheet falls much slower than the crumpled ball, which they note as an air-resistance effect.	you hear one thud or two?' Move among groups asking: 'Which object do you expect to take longest? Does your table show that?' When groups finish, collect the means on the board with the reporter from each group. Say: 'Notice the numbers: 0.63, 0.64, 0.65. Why are they so close when a stone is far heavier than paper?' Wait 10-15 seconds, then cold-call 3 learners. Ask: 'Why did the flat paper fall slowly? What is different about the crumpled ball?' Enforce safety: 'No hands or feet under the drop.' Point out that this is the same gravity that pulls Mzee Kamau's cart: 'On the slope it pulls along the road. Here it pulls straight down.'		readiness. Look for learners who use air resistance to explain the flat sheet.
Explain Phase  VIDEO: Acceleration due to gravity at the space station <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Acceleration Due to Gravity (Flexbook) Source: CK-12  Search ARES for similar readings	(About 20 minutes) Learners use the class data to explain what they saw and to calculate g. Step 1: They state a claim in their journals: 'All three objects reached the ground at about the same time, so gravity gives them the same acceleration (mass does not matter) when air resistance is negligible.' Step 2: Using Lesson 2's $s = \frac{1}{2}at^2$ with $u = 0$ and the sign convention from Lesson 3 (down taken as positive, so $a = +g$), they rearrange to $g = 2h/t^2$. Step 3: They substitute the class mean: $g = 2 \times 2.0 / 0.64^2 = 4 / 0.4096 = 9.8 \text{ m/s}^2$ (2 s.f.). Step 4: They compare with theory using $g = 10 \text{ m/s}^2$: $t = \sqrt{2h/g} = \sqrt{0.4} = 0.63 \text{ s}$, which is within reaction-time error of 0.64 s. They are told that calculations in this course use $g = 10 \text{ m/s}^2$, and that 9.8 m/s^2 is the measured value. Step 5: They connect to the phenomenon: every second in free fall adds about 10	Say: 'Here is the claim we will test with the evidence: all objects fall with the same acceleration when air resistance is negligible.' Ask learners to write the claim and the evidence from the table. Cold-call 3 learners to state the evidence aloud. Write $s = \frac{1}{2}at^2$ on the board and ask: 'Our object starts from rest, so $u = 0$. If a is g and s is h, what does the equation become?' Wait 10-15 seconds. Cold-call 2 learners and write $g = 2h/t^2$. Walk through the substitution in steps on the board, with units, and have learners do the arithmetic in pairs ($4 / 0.4096$). Ask: 'Our g is 9.8 and the textbook uses 10. Is our experiment wrong?' Wait 10-15 seconds, then cold-call 3 learners to discuss reaction time, rounding and air resistance. Say: 'In this course we use $g = 10 \text{ m/s}^2$, so exercises are clean.' Ask: 'The mass of the cart was 300 kg. Did that change how it accelerated?	Claim-Evidence-Reasoning (CER): learners state the claim, cite their class data and justify with the equation $s = \frac{1}{2}at^2$, so the conclusion is built from evidence and mathematics instead of authority. Reusing earlier equations (Lesson 2) shows how a measurement turns into a physical constant.	Check journals for: (a) a claim that matches the evidence, (b) citations of the numbers (0.63, 0.64, 0.65 s), (c) correct rearrangement $g = 2h/t^2$, and (d) a calculated g near 9.8 m/s^2 with units. Indicators: learners explain that differences are within reaction-time error; learners say that the mass of the object does not change the acceleration; learners do not say that heavier objects fall faster. Ask 3 learners to calculate g for a single time (for example, 0.65 s gives 9.5 m/s^2) on mini-whiteboards and hold them up for a quick scan.

	m/s to the speed, so a vertical drop of 20 m takes only 2 s, whereas the cart on the gentle escarpment slope had to travel 100 m. They write one sentence on why the mkokoteni's acceleration is smaller than g (the slope reduces the pull along the road) as a prediction to be examined in Lesson 5.	What does today's evidence suggest?' Wait 10-15 seconds and cold-call 2 learners. Link to the road: 'On the cliff the pull is the whole of g. On the escarpment, only part of it.' Do not teach the 0.2 fraction calculation (Lesson 5).		
Driving Question Board (DQB) Creation  VIDEO: Viewing g as the value of Earth's gravitational field near the surface <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Acceleration Due to Gravity Practice Source: CK-12  Search ARES for similar readings	(About 8 minutes) Learners return to the class Driving Question Board started in Lesson 1: 'Why does a runaway mkokoteni on the Limuru escarpment keep getting faster, and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?' On sticky notes, each learner writes one new question or partial answer from today's evidence, for example: 'If gravity gives every object the same acceleration, why does the cart accelerate at only 2 m/s²?' 'Why did the flat paper fall slowly?' 'How fast would an object be moving after 2 s in free fall?' 'Could we use free fall to predict how far something drops?' They place notes under headings: Answered, Partly answered and New questions. Learners also move the Lesson 4 notes on gravity to the 'Partly answered' section of the DQB, marking what is known (gravity accelerates all objects equally; g is about 10 m/s²) and what is not yet known (how the slope changes the acceleration).	Say: 'Take one sticky note. Write a question that today's drops make you wonder about, or an answer you can now give to a question already on the board.' Give 2 minutes of silent writing. Invite 4 learners to read their notes and place them. Ask: 'Which question on the board does today's evidence help answer?' Wait 10-15 seconds. Cold-call 2 learners. Point to any question about why the cart's acceleration is not as big as g and say: 'This is a question we will answer in Lesson 5.' Highlight questions about air resistance and say: 'Good, we ignore it for now but we must keep it in mind.' Move answered notes into the correct column and thank learners for the question.	Driving Question Board: a public record that shows how each piece of evidence moves the class from puzzlement to explanation, giving learners ownership of the questions and signalling what has yet to be explained.	Check sticky notes for questions that link free fall to the mkokoteni, the slope or air resistance. Indicators: at least 80 percent of learners submit a question or answer; notes refer to evidence (for example '0.64 s') instead of vague statements; some learners link to Lesson 2's equation or Lesson 3's sign convention. Notice learners who still ask whether heavier objects fall faster; follow up with them individually.
Model Building Phase  VIDEO: 02 Freely Falling	(About 7 minutes) Learners open their draft model for the cart, an annotated diagram of the cart on the escarpment, which in Lesson 7 will be completed with a velocity-	Say: 'Take out your model. Add a panel for free fall.' Show an example on the board: an arrow down, $a = g$, and the note about mass. Ask: 'Where on your model	Cumulative model building: adding each new piece of evidence to one visual model lets learners integrate ideas across lessons and notice gaps (here, how slope affects	Check models for: (a) a free fall panel with $g = 10 \text{ m/s}^2$ and the equation $g = 2h/t^2$, (b) a statement that mass does not change acceleration, (c) a sensible

Objects And Acceleration Due To Gravity Source: Seavuria  Search ARES for similar videos  HTML: Acceleration Due to Gravity (Flexbook) Source: CK-12  Search ARES for similar readings	time graph. They add a 'Free fall evidence' panel. It has a vertical arrow showing an object dropped from 2.0 m, the label ' $a = g = 10 \text{ m/s}^2$ downward (measured 9.8 m/s^2)', the relation $g = 2h/t^2$, and the note 'mass does not change the acceleration when air resistance is negligible'. Beside the cart drawing they add a question mark note: 'The slope gives a smaller acceleration than g . How much smaller?' They also add a mini table of the cart's first rows from earlier lessons ($t = 1 \text{ s}$: $v = 2 \text{ m/s}$, $s = 1 \text{ m}$) next to the idea that free fall would give $v = 10 \text{ m/s}$ after 1 s, as a prediction for comparison. They reflect in one sentence: 'Today I learned that ____, which helps explain the mkokoteni because ____.'	does gravity act on the cart? In which direction?' Wait 10-15 seconds, then cold-call 2 learners. Say: 'The cart is not in free fall. What do you think is different?' Take 3 responses and accept ideas about the slope without confirming. Say: 'In the next lesson we will use numbers to test your idea.' Circulate and check that each model uses units and the sign convention (down positive). Remind learners to keep models for Lesson 7.	acceleration), which maintains coherence in the storyline.	question about the slope, and (d) units on all numbers. Indicators of understanding: the learner connects gravity to the cart's steady increase in velocity; the learner does not claim the cart accelerates at 10 m/s^2 . Use the reflection sentence as an exit ticket and sort it into 'got it', 'nearly' and 'needs support' for follow-up in Lesson 5.
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D. TEACHER REFLECTION

After the lesson consider: (1) How many learners changed their prediction after seeing the three class means of 0.63, 0.64 and 0.65 s, and which reasons did the learners who did not change give? (2) Did groups control the 2.0 m height and release method, and how reliable were their stopwatch times? (3) Could learners explain why 9.8 m/s^2 and 10 m/s^2 are both acceptable, and did anyone treat the difference as an error in the experiment? (4) Did learners confuse the stone and the crumpled paper ball results with the flat sheet and air resistance? (5) Did learners apply $s = \frac{1}{2}at^2$ with $u = 0$ correctly and use the sign convention from Lesson 3? (6) What misconceptions about heavier objects remain, and how will you address them in Lesson 5? (7) Were safety rules followed during dropping? Record any adjustments to apparatus (for example, a phone slow-motion video for timing) for future lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We dropped a rubber bung, a stone and a crumpled paper ball from 2.0 m. Mean fall times were 0.63 s, 0.64 s and 0.65 s (class mean 0.64 s), almost the same although the masses were very different. A flat sheet of paper fell more slowly than the crumpled ball.
What did I learn?	When air resistance is negligible, all objects fall with the same acceleration due to gravity, regardless of mass. Using $s = \frac{1}{2}at^2$ with $u = 0$, $g = 2h/t^2 = 4/0.64^2 = 9.8 \text{ m/s}^2$, and we use $g = 10 \text{ m/s}^2$ in calculations. Differences are within reaction-time error.
How does this explain the phenomenon?	Gravity pulls on the mkokoteni all the time, so it keeps getting faster, and free fall shows what gravity does on a vertical path: every second adds about 10 m/s to the speed of any object. The cart on the escarpment is not in free fall, so its acceleration is smaller than g (to be investigated in Lesson 5), but the same equations apply to predict its speed and distance.

LESSON 5 (80 minutes (2 x 40-minute lessons)): Freely Falling Bodies: Explain and Calculate

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the equations of linear motion with $u = 0$ and $a = g = 10 \text{ m/s}^2$ to free-fall problems, and explain why the slope reduces the mkokoteni's acceleration to $a = g \times 0.2 = 2.0 \text{ m/s}^2$, so that the cart and a free-falling body can reach the same 20 m/s in very different times.
Knowledge	By the end of the lesson, the learner should be able to: (a) calculate variables of motion using equations of linear motion with constant acceleration ($v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$; $s = \frac{1}{2}(u+v)t$); (b) investigate the effect of gravity on bodies under free fall, using the class data from Lesson 4 (g estimated as 9.8 m/s^2 , exercises use $g = 10 \text{ m/s}^2$); (c) appreciate the significance of linear motion and free fall in real life. Learners will also state that the mkokoteni is NOT in free fall: its acceleration along the slope is $g \times 0.2 = 2 \text{ m/s}^2$, while free fall on a vertical path has $a = g = 10 \text{ m/s}^2$.
Skills	Critical thinking and problem solving (listing knowns and unknowns, choosing an equation, using the sign convention); communication and collaboration (explaining calculation steps to peers); learning to learn (reflecting on the Lesson 4 data and on predictions). NGSS practices: Using Mathematics and Computational Thinking; Analyzing and Interpreting Data; Constructing Explanations; Developing and Using Models.
Attitudes	Values: Peace (displaying tolerance and calmness when discussing applications of linear motion in real life); Unity (cooperating with others while working on the calculations). PCIs: Road safety (increased awareness of how to minimise fatal injuries resulting from road crashes on sloping roads using the concept of free fall of objects); Life Skills Education (self-awareness) (observing safety measures when objects are dropped). Learners show respect for peers' reasoning and appreciate how mathematics lets us predict danger before it happens.
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	Lesson 4 gave the class a measured g (9.8 m/s^2 from $h = 2.0 \text{ m}$ and a mean time of 0.64 s). Lesson 5 turns that evidence into prediction: learners use $u = 0$ and $a = g$ to calculate for a mango, then a 45 m drop, and then compare free fall with the mkokoteni on the escarpment. The key insight is that the cart reaches the same 20 m/s as a 20 m free fall but needs 10 s instead of 2 s , because the slope supplies only 0.2 of g . This links to the DQB and sets up Lesson 6 (stopping distances) and Lesson 7 (velocity-time graph).
Safety Notes	This is a calculation lesson; no drops are required. If learners re-demonstrate any drop for the DQB, drop only light objects (rubber bung, paper ball) from no more than 2 m , with the area below cleared, and nobody standing underneath. Learners must NOT drop anything from a window, a building or a tree, and must not climb anything. No real mkokoteni or vehicle is to be released on any slope; the escarpment scenario is worked on paper only. Keep feet clear of the demonstration area and report any accident to the teacher at once.

B. LESSON OVERVIEW

Learners recall the Mzee Kamau scenario: a loaded mkokoteni (300 kg) released from rest at the top of a steep 100 m section of the Limuru-Nairobi escarpment road crashes into a parked lorry at 20 m/s (72 km/h) after 10 s , and Achieng says 'it started slowly then flew'. They first predict the mango fall time and the drop height for a 30 m/s impact, then review the Lesson 4 data ($g = 9.8 \text{ m/s}^2$ estimated; $g = 10 \text{ m/s}^2$ used for calculation). In the Observe phase they work through three calculations as evidence: (a) a mango falling from a 5 m branch in a school compound ($t = 1 \text{ s}$, $v = 10 \text{ m/s}$); (b) an object hitting the ground at 30 m/s (dropped from 45 m , falling for 3 s); (c) the cart at $s = 25 \text{ m}$, using $a = 2$ and not 10 ($v = 10 \text{ m/s}$, $t = 5 \text{ s}$). In the Explain phase they compare a 20 m free fall (2 s , 20 m/s) with the cart (10 s , 20 m/s) and explain that the slope gives only 0.2 of g , and that free fall is the extreme case of a vertical slope. They then discuss dangers on steep roads and ways to reduce them, update the DQB and revise the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: 02 Freely Falling Objects And Acceleration Due To Gravity Source: Seavuria  Search ARES for similar videos  HTML: Acceleration Due to Gravity (Flexbook) Source: CK-12  Search ARES for similar readings	Learners sit in groups of four. They look at the sketch of the escarpment (100 m road, 20 m vertical drop, lorry at the bottom) and the sketch of a mango tree in a school compound with a 5 m branch. Individually, in their notebooks, they write predictions for three questions: (1) How long will a mango take to fall from the 5 m branch, and how fast will it be going when it lands? (2) An object hits the ground at 30 m/s: from what height was it dropped? (3) If the cart fell vertically through 20 m instead of rolling down the slope, would it reach the bottom sooner or later than 10 s, and would its final speed be bigger, smaller or the same as 20 m/s? They give a reason for each prediction and rate their confidence (1-3). Groups then share predictions and sketch them as quick estimates on a sheet of paper. Predictions link to the phenomenon: the cart gets faster because gravity keeps pulling it down the slope.	Display the escarpment sketch and say: 'Remember Mzee Kamau's cart. Achieng said it started slowly then flew. Today we turn Lesson 4's measured g into predictions.' Give 3 minutes of silent writing: 'Write your own prediction first. No talking yet.' Then say: 'Before we calculate anything, what do you think?' WAIT TIME 10-15 seconds before taking answers. Cold-call 4 learners (from different groups) for question 1, and 3 learners for question 3, asking each: 'What is your reason?' Do NOT confirm or correct. Record the class predictions in a tally on the board (sooner / later / same). Say: 'Keep your predictions. We will test them with numbers.' Circulate and note common ideas such as 'heavier things fall faster' or 'free fall is the same as the cart'.	Predict-with-reasons (eliciting prior ideas): writing a prediction and a reason before any calculation makes learners commit to an idea, so that later evidence either supports it or creates productive conflict. The confidence rating lets learners see how their certainty changes with evidence.	Observable indicators: (1) every learner has written a prediction and a reason for all three questions; (2) teacher tallies how many predict 'later' for the free-fall time compared with the cart (the correct idea) vs 'sooner' or 'same'; (3) teacher listens for the misconception that the cart's mass or load of maize changes its acceleration, or that the cart is in free fall; (4) at least 4 different learners voice reasoning in the cold calls.
Observe Phase  VIDEO: 02 Freely Falling Objects And Acceleration Due To Gravity Source: Seavuria  Search ARES for similar videos  EXERCISE: Acceleration Due to Gravity	Learners gather evidence in two steps. Step 1: Recall the Lesson 4 data ($h = 2.0$ m; mean times: rubber bung 0.63 s, stone 0.64 s, paper ball 0.65 s; mean 0.64 s). They recompute $g = 2h/t^2 = 4/0.64^2 = 4/0.4096 = 9.8$ m/s ² and note that theory with $g = 10$ gives $t = \sqrt{0.4} = 0.63$ s, and that the differences are within reaction-time error. They agree to use $g = 10$ m/s ² in the exercises. Step 2: In groups, they work through three	Give out worksheets. Say: 'First, list what you know and what you want. Which equation has exactly those variables?' Remind learners that Lesson 2 taught choosing an equation from the knowns and Lesson 3 taught the sign convention. Write on the board: 'List u, v, a, s, t. Mark the unknown. Pick the equation.' Circulate with these prompts: 'Why is $u = 0$ here?' 'Why $a = 10$ for the mango but $a = 2$ for the cart?'	Data-based reasoning with a structured problem-solving routine (known-unknown-equation-check): using the same steps for every problem lets learners focus on the physics, not on guessing. Comparing the cart's calculation with the mkokoteni table (a data check) builds confidence that the equations predict real motion, and the contrast between $a = 10$ and $a = 2$ exposes the role of the slope.	Observable indicators: (1) learners list knowns and unknowns before computing; (2) correct answers: (a) $t = 1$ s and $v = 10$ m/s; (b) $h = 45$ m and $t = 3$ s; (c) $v = 10$ m/s and $t = 5$ s, with units; (3) learners use $a = 2$ for the cart and not 10; (4) learners can say that mass does not appear in the equations for free fall and does not change the cart's acceleration; (5) teacher notes which groups confuse $g = 9.8$ and $g = 10$ and reminds them

Practice Source: CK-12 Search ARES for similar readings	calculations on a worksheet, writing known values (u, a, s, v, t), choosing an equation and giving units. Down is positive. (a) Mango from a 5 m branch: $u = 0$, $a = 10$, $s = 5$ so $s = \frac{1}{2}at^2$ gives $5 = 5t^2$, $t = 1$ s, then $v = u + at = 10$ m/s. (b) Object hits the ground at 30 m/s: $u = 0$, $v = 30$, $a = 10$, so $s = v^2/2g = 900/20 = 45$ m, and $t = v/a = 3$ s. (c) The cart at $s = 25$ m on the slope: $u = 0$, $a = 2$ (NOT 10), $v^2 = 2as = 2 \times 2 \times 25 = 100$, so $v = 10$ m/s, and $t = v/a = 5$ s. They check (c) against the mkokoteni table ($t = 5$ s: $v = 10$ m/s, $s = 25$ m). Each group fills in a results table and cross-checks answers with a neighbouring group.	'What are your units?' For calculation (c), stop the class and ask: 'The cart weighs 300 kg. The mango is much lighter. Does the cart accelerate faster?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Prompt: 'Does mass appear in any of the four equations?' Make sure every group finishes (a) before (b), and (b) before (c). Do not give answers; let groups check each other's results. After 20 minutes, record each group's answers on the board and mark any disagreement for discussion.		that exercises use $g = 10 \text{ m/s}^2$.
Explain Phase  VIDEO: Acceleration due to gravity at the space station <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Acceleration Due to Gravity (Flexbook) Source: CK-12 Search ARES for similar readings	Learners use their results to explain the phenomenon. Task 1 (free fall vs cart): they calculate a 20 m free fall: $u = 0$, $a = 10$, $s = 20$, so $t = \sqrt{2 \times 20/10} = 2$ s and $v = 10 \times 2 = 20$ m/s. They put it side by side with the cart (10 s, 20 m/s) in a two-column table, and find that the final speeds are the same (20 m/s = 72 km/h) while the times differ (2 s vs 10 s). Task 2 (why the slope matters): they compute the ratio of accelerations $2/10 = 0.2$ and relate it to $\sin(\text{angle}) = 0.2$, i.e. the vertical drop of 0.2 m per metre travelled; they write in their own words: 'The slope gives the cart only 0.2 of g. A vertical path ($\sin = 1$) gives $a = g$, so free fall is the extreme case of a slope.' Task 3 (explaining Achieng's 'started slowly then flew'): they use $v = 2t$ and $s = t^2$ to explain that velocity begins at 0 and rises by 2 m/s every second, and that the distance grows as t^2 (25 m by 5 s, but 100 m by 10 s, so 75 m in the last 5 s). Task 4 (dangers and	Say: 'We have two motions that end at the same speed. Why does one take 2 seconds and the other 10?' WAIT TIME 10-15 seconds and take no hands. Cold-call 3 learners one after another, each adding to the previous answer: 'Add to what she said.' Push for evidence: 'Which number shows that? Where is it in your table?' Ask: 'If the slope were twice as steep, what happens to a? What if it were vertical?' Draw an escarpment triangle and a vertical line on the board. Challenge the idea that the cart is in free fall: 'The cart has no engine and nobody pushes it. Is it in free fall? Why not?' (Free fall is motion under gravity alone along a vertical path with negligible air resistance; the cart is on a slope and the road holds part of the pull.) Use the road-safety discussion: 'Mzee Kamau's cart reached 72 km/h in 10 s. What could reduce the danger?' Take answers from 4 groups and press them to link	Claim-Evidence-Reasoning with a compare-and-contrast table: learners state a claim ('the slope reduces the cart's acceleration to 0.2 g'), cite numbers (2 s vs 10 s, 20 m/s vs 20 m/s) and give the reasoning (only part of gravity acts along the slope; a vertical path gives full g). Contrasting two cases of the same physics shows that free fall and the cart follow one set of equations, differing only in a.	Observable indicators: (1) learners correctly state $t = 2$ s and $v = 20$ m/s for the 20 m free fall; (2) learners explain that the final speed is the same but the time is different because the slope gives $a = 2 \text{ m/s}^2$, which is 0.2 of g; (3) learners say the cart is not in free fall; (4) learners link 'started slowly then flew' to $v = 2t$ and $s = t^2$ using numbers (e.g. 25 m at 5 s, 100 m at 10 s); (5) learners name at least one real-life importance of free fall and one way of reducing danger; (6) at least 6 different learners speak during the discussion.

	mitigation): in groups they discuss, for Key Inquiry 1 and 2, where free fall matters in real life (a mango falling, objects falling from scaffolding at a Nairobi construction site, a parked vehicle rolling downhill) and how dangers are reduced (escape lanes, warning signs on steep roads such as the Kinangop descent, wheel chocks, handbrakes, hard hats). Each group presents one claim with evidence from the calculations.	each safety measure to the physics (e.g. a wheel chock stops acceleration from starting; an escape lane gives a long distance to stop). Do not teach stopping distance in detail; say: 'Lesson 6 will measure stopping distances.' Do not re-teach graphs; Lesson 7 covers them.		
Driving Question Board (DQB) Creation  VIDEO: Viewing g as the value of Earth's gravitational field near the surface <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Acceleration Due to Gravity Practice Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board started in Lesson 1. In groups, they (1) place sticky notes with answers for questions that can now be answered (e.g. 'Why does the cart keep getting faster with no one pushing?' - Gravity pulls it down the slope all the time, so $a = 2 \text{ m/s}^2$ and v rises 2 m/s each second; 'Could we have predicted the speed?' - Yes, $v = 2t$ gives 20 m/s at 10 s and $v^2 = 2as$ gives 20 m/s at 100 m); (2) write at least one new question arising from today's work, such as 'Why does doubling speed make stopping so much harder?', 'What does the graph of the cart's motion look like?' or 'Does air resistance matter when a mango falls?'; and (3) move any answered question to the 'Answered' column, writing the evidence (calculation) on the note. They mark questions still open for later lessons.	Say: 'Look at the board. Which questions can we now answer with numbers, not just opinions?' WAIT TIME 10-15 seconds. Invite 3 groups to place and defend one answered note, asking 'What evidence supports it?' Say: 'Write one new question that today's work made you ask.' Cold-call 3 learners to read new questions aloud. Sort new questions: stopping distance and speed ones are linked to Lesson 6; graph questions are linked to Lesson 7. Do not answer them now. Point out any question about whether mass changes the cart's acceleration and ask learners to cite the calculation for the answer. Photograph or keep the DQB for the next lesson.	Driving Question Board as a public record of growing understanding: moving questions from 'open' to 'answered' with evidence shows progress and motivates the next lessons, while new questions keep learning driven by the phenomenon and by learners' own curiosity.	Observable indicators: (1) each group moves at least one question to 'Answered' and writes evidence with numbers; (2) each group writes at least one new, relevant question; (3) learners use terms such as acceleration, free fall, g and the equations correctly; (4) teacher notes which learners still ask about mass or 'weight making it faster' and plans a quick follow-up.
Model Building Phase  VIDEO: 02 Freely Falling Objects And Acceleration Due	Learners revise the class model of the mkokoteni (an annotated diagram; the velocity-time graph is built in Lesson 7). On the diagram of the 100 m escarpment road they (1) label the slope ($\sin = 0.2$; 20 m vertical drop), the force of gravity,	Say: 'Add today's evidence to your model. Where does g appear? Where does 0.2 appear?' Circulate and ask: 'How does your model show why the cart starts slowly?' 'How does it show that free fall is different from the cart?' WAIT	Cumulative model revision: learners add evidence and mathematical relationships to an annotated diagram, and so connect observations (the cart gets faster), mechanism (gravity along the slope, $a = g \times 0.2$) and	Observable indicators: (1) the model shows $a = 2.0 \text{ m/s}^2$ for the cart and $a = g = 10 \text{ m/s}^2$ for free fall, with no confusion between them; (2) speeds and distances at $2, 5$ and 10 s match the table ($4, 10, 20 \text{ m/s}$; $4, 25, 100 \text{ m}$); (3) the

To Gravity Source: Seavuria  Search ARES for similar videos  HTML: Acceleration Due to Gravity (Flexbook) Source: CK-12  Search ARES for similar readings	the direction of motion and $a = 2.0 \text{ m/s}^2$; (2) add speeds at $t = 0, 2, 5$ and 10 s ($0, 4, 10, 20 \text{ m/s}$) and distances ($0, 4, 25, 100 \text{ m}$) from the table; (3) add a small second sketch of the 20 m vertical free fall with $a = g = 10 \text{ m/s}^2$, $t = 2 \text{ s}$, $v = 20 \text{ m/s}$; (4) write a caption: 'Free fall is the extreme case of a vertical slope; the slope reduces a to $0.2g$, so the cart takes 10 s but reaches the same 20 m/s '; and (5) write a safety note linking speed to danger on steep roads. They compare their revised model with their Lesson 1 model and note what has changed.	TIME 10-15 seconds after each question. Cold-call 3 learners to explain one revision and the reason for it. Ask 2 pairs to swap models and give one piece of 'two stars and a wish' feedback. Use the check: 'If $a = 2$, what is v at 3 s ?' (6 m/s) Collect models and note gaps (e.g. use of $a = 10$ for the cart). Finish by asking each learner to complete the summary table in the notebook: what we observed, what we learned, how we can explain it.	prediction (equations of linear motion). Comparing with the earlier model makes learning visible and builds the explanation for the final synthesis in Lesson 8.	caption explains that the slope gives $0.2g$; (4) the model includes a link to danger and road safety; (5) learners can explain one revision to the model and justify it with a calculation.
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D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners choose equations by listing knowns and unknowns, or did they guess? Which groups needed help? (2) How many learners still think the cart is in free fall or that its 300 kg load changes its acceleration? What evidence from the calculations changed their minds? (3) Did learners distinguish $g = 9.8 \text{ m/s}^2$ (measured in Lesson 4) from $g = 10 \text{ m/s}^2$ (used in exercises) and explain why? (4) Was the comparison of 2 s and 10 s with the same 20 m/s clear? Which wording or sketch helped most? (5) Did the road-safety discussion connect to the physics and to local roads (Limuru, Kinangop descent), and were all learners able to contribute? (6) Which new DQB questions should be used in Lesson 6 (stopping distance) and Lesson 7 (graphs)? What adjustments are needed to the timing, grouping or worksheet?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe and calculate today? (Include: the mango from the 5 m branch takes $t = 1 \text{ s}$ and lands at $v = 10 \text{ m/s}$; an object hitting the ground at 30 m/s was dropped from 45 m and fell for 3 s ; the cart at $s = 25 \text{ m}$ has $v = 10 \text{ m/s}$ after $t = 5 \text{ s}$; a 20 m free fall takes 2 s and gives 20 m/s while the cart takes 10 s for the same 20 m/s .)
What did I learn?	What did we learn? (Include: with $u = 0$ and $a = g = 10 \text{ m/s}^2$, the equations of linear motion predict time, speed and height for freely falling bodies; mass does not change the acceleration; the mkokoteni is not in free fall because on the slope its acceleration is $g \times 0.2 = 2 \text{ m/s}^2$; free fall is the extreme case of a vertical slope.)
How does this explain the phenomenon?	How can we now explain the phenomenon? (Include: the cart keeps getting faster because gravity pulls it down the slope all the time, so its velocity rises by 2 m/s every second; it starts slowly because velocity begins at zero and feels sudden because distance grows as t^2 ; the same equations predict 20 m/s (72 km/h) after 10 s and 100 m , and show why steep roads are dangerous.)

LESSON 6 (80 minutes (2 x 40-minute lessons)): Road Safety: Stopping Distance and Real-Life Applications of Linear Motion

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use the equations of linear motion to explain why a runaway mkokoteni hits a lorry at 20 m/s, and why higher speeds on roads like Mombasa Road need far longer stopping distances. They connect the physics to NTSA road-safety rules and to one further application of linear motion in Kenya.
Knowledge	By the end of the lesson, the learner should be able to: (a) explain that stopping distance = reaction distance + braking distance; (b) calculate braking distance using $v^2 = u^2 + 2as$ with $v = 0$ and $a = -5 \text{ m/s}^2$ (dry) or -2.5 m/s^2 (wet); (c) show that braking distance depends on the square of speed (braking distance = $u^2/2 a $), so doubling the speed multiplies it by 4 and tripling it by 9; (d) discuss the dangers of accelerating on steeply sloping roads and the safety measures used to reduce fatal injuries; (e) appreciate the significance of linear motion in real life.
Skills	Calculating with the equations of linear motion (NGSS: Using Mathematics and Computational Thinking); analysing and interpreting data from a table (NGSS: Analyzing and Interpreting Data); constructing explanations from evidence (NGSS: Constructing Explanations); researching with digital or print media and presenting findings in a group (Communication and collaboration).
Attitudes	Learners value road safety and take responsibility for safe behaviour as passengers, pedestrians and drivers. Peace: they stay calm and tolerant when discussing applications. Unity: they cooperate in group research. Respect: they value others' opinions when debating road-safety rules.
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	Lessons 1 to 5 showed that the mkokoteni accelerates at a constant 2.0 m/s^2 and reaches 20 m/s (72 km/h) after 10 s and 100 m, and that free fall is the same physics with $a = g$. Lesson 6 turns this knowledge to the other side of the problem: how far does a vehicle travel while stopping? It moves the driving question from predicting speed and distance to using that prediction to save lives. Learners see that speed matters most because distance depends on v^2 . Lesson 7 will then bring everything together on the velocity-time graph model.
Safety Notes	Do the stopping-distance activity on the school field or in the corridor, never on a real road. Use chalk or string for the marked distances, and keep walking-pace trials away from other learners. Keep the cart demonstration to a small toy or trolley on a short plank, with a box to catch it. Learners using digital devices must follow the school rules on safe use and online safety. Treat discussion of road crashes sensitively, since some learners may have been affected by one.

B. LESSON OVERVIEW

Learners begin by predicting how far a matatu on Mombasa Road needs to stop when it is travelling at 36, 72 and 108 km/h, and whether doubling the speed doubles the distance. In the Observe Phase they find the answer in a data table. They convert the speeds to 10, 20 and 30 m/s, calculate reaction distance (speed x 1.0 s) and braking distance ($v^2 = u^2 + 2as$ with $v = 0$, $a = -5 \text{ m/s}^2$ dry and -2.5 m/s^2 wet), and compare the results. In the Explain Phase they link the results to the mkokoteni, which has no brakes and ends at 20 m/s, to the dangers of speed on slopes, and to Kinangop descent signs and NTSA guidelines. They then research one other Kenyan application of linear motion (SGR train, pole vault or construction safety) for a presentation in Lesson 7. The lesson ends by updating the DQB and the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: linear motion stopping distance  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings Search terms: linear motion stopping distance	(10 min) Learners recall the mkokoteni crashing into the parked lorry at 20 m/s (72 km/h) and are shown a sketch of a matatu on Mombasa Road with a boy stepping onto the road ahead. Individually they answer on paper: (1) What two things happen between the driver seeing danger and the matatu stopping? (2) If the matatu is at 36 km/h and needs about 20 m to stop, how far will it need at 72 km/h and at 108 km/h? They write a number and a reason for each, then sketch a quick bar or distance line for the three speeds. They then share with a partner and note where they disagree. Most will predict 40 m and 60 m (a doubling and a tripling), a useful misconception to test.	Say: "Last lesson, our mkokoteni had no brakes and no driver. Today the matatu has both. Think about Mombasa Road. Between the driver seeing danger and the matatu coming to a stop, what happens?" WAIT TIME 10-15 seconds in silence before anyone speaks. Then say: "Write your prediction for 72 km/h and 108 km/h. Do not worry about being right; I want your reasoning." Walk around and cold-call 4 learners to give their numbers and reasons, without saying who is right. Record the predictions on the board in a table headed "Prediction before evidence" so they can be compared later. Ask: "Who predicted the distance doubles when the speed doubles? Hands up. Who thinks it might be more than double?" Count hands and write the counts down. Do not correct anyone; say: "We will test that with evidence in a moment."	Predict-Observe-Explain (prediction stage). Committing to a written prediction and reason makes learners aware of their own idea (distance is proportional to speed), so the evidence later produces a real cognitive conflict instead of being passively received. Cold-calling shows learners that the class holds several different ideas.	Check the written predictions. Look for: (a) whether learners identify two stages (reaction time and braking); (b) whether they predict a linear pattern (40 m, 60 m) or a faster growth; (c) whether they link the distance to the speed using ideas from Lesson 3. Tally the hand counts. Note learners who give no reasoning and return to them in the Observe Phase.
Observe Phase  VIDEO: No closely matching video in the ARES library for this activity. Search ARES for videos Search terms: linear motion stopping distance  READING: No closely matching reading in the ARES library for this activity. Search ARES for readings	(25 min) In groups of 4, learners get the Stopping Distance Data Sheet for a matatu on Mombasa Road (driver reaction time 1.0 s; braking deceleration 5 m/s ² on dry road and 2.5 m/s ² on wet road). Part A (10 min): they convert 36, 72 and 108 km/h to m/s by dividing by 3.6 (10, 20, 30 m/s). Reaction distance = speed x 1.0 s gives 10 m, 20 m and 30 m. Part B (10 min): for braking distance they use equation (3), $v^2 = u^2 + 2as$, with $v = 0$ and $a = -5 \text{ m/s}^2$: $0 = u^2 - 10s$, so $s = u^2/10$. Dry road: 10 m, 40 m, 90 m. Wet road with $a = -2.5 \text{ m/s}^2$: 20 m, 80 m, 180 m. They complete the table: total stopping distance on the dry road is 20 m at 10 m/s,	Hand out the data sheets and say: "Convert first. Dividing by 3.6 gives metres per second. Check with your partner before you go on." Circulate and look for the common mistake of using $a = +5$ instead of -5 or forgetting that $v = 0$. Say: "Which equation has no time in it? Why is that the one we want?" WAIT TIME 10-15 seconds, then cold-call 3 groups to explain their choice of $v^2 = u^2 + 2as$. Say: "Look at the 72 km/h row. The speed is double the first row. What happened to the reaction distance? What happened to the braking distance?" Cold-call 4 learners for their results. Do not give the pattern; ask: "Look down	Data table analysis with pattern finding. Learners produce the data themselves from the equations of linear motion and then look for relationships (reaction distance is proportional to speed; braking distance is proportional to speed squared). Calculating the numbers first and then spotting the pattern builds a deeper understanding than being told that distance goes with v^2 .	Observable indicators: (a) learners write $a = -5 \text{ m/s}^2$ as negative; (b) they use $v = 0$ in $v^2 = u^2 + 2as$; (c) the braking distances are 10, 40, 90 m (dry) and 20, 80, 180 m (wet); (d) totals are 20, 60, 120 m for the dry road; (e) learners can state that the braking distance is 4 times as much when the speed doubles. Use a thumbs up/sideways/down check: "Can you get the braking distance at 20 m/s on your own?" Pair a learner who is struggling with one who has completed the table.

Search terms: linear motion stopping distance	60 m at 20 m/s and 120 m at 30 m/s. Part C (5 min): they compare their numbers with their predictions and look for a pattern in the braking column, e.g. 10 to 40 to 90 (x1, x4, x9). Practical option on the school field: learners mark out 10 m, 20 m and 30 m with chalk and walk them at steady pace, counting 1 second per 'step' to feel what 'reaction distance' means.	the braking column: 10, 40, 90. What is the pattern?" WAIT TIME 15 seconds. If learners are stuck, prompt: "Compare it with 1, 2, 3 for the speeds." Remind them to keep the units (m, s, m/s, m/s²) in every line. Check that every group has the reaction distance, braking distance and total for both road conditions.		
Explain Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i>  Search ARES for videos Search terms: linear motion stopping distance  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: linear motion stopping distance	(25 min) Part A (10 min): Whole-class discussion. Learners compare the results with their predictions and explain in their own words, using evidence from the table: (1) why the total stopping distance is more than double when the speed doubles (20 m to 60 m); (2) why the wet road doubles the braking distance ($a = -2.5$ instead of -5); (3) what happens at 108 km/h on a wet road (30 m reaction + 180 m braking = 210 m). Learners write a claim-evidence-reasoning statement: "Braking distance depends on the square of speed because $v^2 = u^2 + 2as$ gives $s = u^2/2 a$; doubling u multiplies s by 4." Part B (10 min): Link to the phenomenon and slopes. In groups, learners compare the cart (no brakes; it reaches 20 m/s after 100 m) with a matatu stopping from 20 m/s (60 m total on a dry road). They discuss: Why are the signs on the Kinangop descent (steep gradient, slow down, use low gear, runaway truck ramps) there? How does the slope make a long descent dangerous, based on what they learned in Lesson 5 about speed and gravity? What do NTSA rules (speed limits, keeping a safe following distance, no overloading) do to reduce the risk?	Say: "Look at your predictions from the start of the lesson. Turn to your partner and tell them one thing that was different from what you expected." Cold-call 4 pairs. Ask: "Why does the braking distance grow so much faster than the speed?" WAIT TIME 15 seconds. If needed, point to the equation and ask: "In $v^2 = u^2 + 2as$, the u is squared. If u doubles, what happens to u^2?" Then ask: "Mzee Kamau's cart reached 20 m/s with no brakes. A matatu going 72 km/h on a dry road needs 60 m to stop. If a driver is at 108 km/h on a wet road, how far is that? Would a boda-boda rider like Otieno be able to stop?" Cold-call 3 learners. Introduce the Kinangop descent: "Why do you think the signs say 'slow down' at the top of the hill, not halfway down?" Link to Lesson 5 (the danger of speed building up on slopes): the speed gained on a slope is gained as the vehicle descends, and brakes then have to remove much more energy. Say: "What do NTSA rules do about these three ideas: reaction time, speed and road conditions?" Take responses from 4 groups. State the key points clearly and have the class say them: "Stopping distance = reaction distance + braking distance.	Claim-Evidence-Reasoning (CER) with phenomenon linking. Learners state a claim (braking distance goes with v^2), cite evidence from their table, and give a scientific reason from the equation. Connecting this back to the cart and to Kinangop shows the same physics at work in a new context, which helps learners transfer ideas. The group research task extends the idea to other contexts.	Observable indicators: (a) learners state that stopping distance is the sum of reaction and braking distance; (b) they explain, with numbers, that doubling the speed multiplies the braking distance by 4 and tripling it by 9; (c) they give a link between slope, speed and danger; (d) they name at least two NTSA-related safety measures with a physics reason; (e) each group chooses a research topic with a stated linear-motion idea. Use the CER sentences as an exit check: collect 5 samples and check that the evidence cites actual values from the table.

	Part C (5 min): Learners are given the research task for Lessons 6-7 (in groups of 4): the SGR train (scheduling and acceleration), pole vault (motion up and down under g) or construction safety (falling objects). Each group picks one and starts planning how to find and present one linear-motion idea with at least one calculation.	Braking distance goes with the square of the speed." Do not re-teach the equations, only apply them. Be sensitive to learners affected by road crashes. Give the research task with a simple rubric: one linear-motion idea, one calculation, one safety or practical message, 3 minutes to present in Lesson 7. Allow digital devices or print media in school.		
Driving Question Board (DQB) Creation  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms: linear motion, stopping distance  READING: <i>No closely matching reading in the ARES library for this activity.</i> Search ARES for readings Search terms: linear motion, stopping distance	(10 min) Learners revisit the Driving Question Board. Each group reads the question cards on the board and sorts them into: answered by evidence in Lessons 1 to 6, partly answered, not yet answered. For the questions that are answered (for example 'Why does the cart keep getting faster?' and 'How far will it go before it crashes?'), they add an answer card with the evidence and the equation used. They also write new question cards raised today, such as 'Why do heavy lorries need longer to stop?', 'How does a runaway truck ramp work?' or 'What would the cart's stopping distance be if the driver braked?' and add them to the board. Learners tag cards that connect to road safety with a coloured dot.	Say: "Come to the board. Take two minutes in your group to read the cards. Which of them can we now answer using stopping distance?" WAIT TIME 10 seconds before they move. Cold-call 3 groups to move a card and justify it by citing evidence ("We moved it because the table showed..."). Ask: "What new questions did today raise? Write one on a new card." Collect at least 4 new cards and read 2 aloud. Point to the learners' question about heavy loads and say: "That is a great question. The cart was 300 kg and its acceleration did not depend on mass. We need to think about whether braking is different." Mark it for Lesson 7 or 8 discussion. Link the key inquiry questions: "How is free fall and speed important in real life? How can the dangers be reduced? Which cards on the board show that?"	Question tracking on the Driving Question Board. Sorting and moving cards makes the growth of class understanding visible and keeps learning tied to the anchoring phenomenon. Writing new questions encourages learners to see science as an ongoing investigation.	Observable indicators: (a) groups move cards to 'answered' only with a stated piece of evidence; (b) at least 4 new, relevant questions are added; (c) some new questions are about road safety or about applying the equations (not only recall). Note learners who contribute to the sorting and those who do not and invite them to speak.
Model Building Phase  VIDEO: <i>No closely matching video in the ARES library for this activity.</i> Search ARES for videos Search terms:	(10 min) Learners revise the class model for the mkokoteni (an annotated diagram of the escarpment road with the lorry at the bottom, plus a velocity-time sketch). They add a new section, 'What if it needed to stop?' showing: (a) the cart reaching 20 m/s at 100 m; (b) a matatu stopping from 20 m/s: 20 m	Say: "Take out your model from Lesson 5. What do we need to add so that it can explain stopping, not only speeding up?" WAIT TIME 15 seconds. Cold-call 3 learners for ideas. Guide them to add the stopping section and check that the numbers match the table (20 m, 40 m, 60 m total for 20 m/s on dry road; 80 m braking on wet	Cumulative model revision. Adding a stopping component to the class model shows learners that the same equations answer a new question, which consolidates the equations and strengthens the link between the phenomenon and real-life safety decisions. Writing for a community audience encourages clear scientific	Observable indicators: (a) the model includes correct numbers (20 m, 40 m, 60 m) and the v^2 relationship; (b) the written sentence links speed, braking distance and safe driving; (c) the learner mentions that the cart has no brakes or reaction time. Exit-slip check: 'If a matatu's speed goes from 20 m/s to 30 m/s, by

linear motion stopping distance  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: linear motion stopping distance	reaction, 40 m braking, total 60 m on a dry road, and 80 m braking on a wet road; (c) a note saying 'braking distance goes with the square of the speed' and the equation $s = u^2/2 a$. In their notebooks, each learner writes a short answer: 'How can we use mathematics to predict how far a vehicle will travel before it stops, and how does that help us to stay safe?' They add a line linking it to the cart: 'The cart has no reaction time and no brakes, so it travels the full 100 m and hits the lorry at 20 m/s (72 km/h).'	road). Say: "Write one sentence that someone from the Kinangop community could understand: why must a driver slow down early on a steep descent?" Collect 3 sentences and read them out. Remind learners that the full velocity-time graph will be drawn in Lesson 7. Point out to the whole class: "Today we moved from predicting how fast to predicting how far to stop. That is the mathematics that keeps people safe." Collect notebooks or exit slips with the sentence for review.	communication.	what factor does the braking distance increase?' (Expected: 2.25 times, from 40 m to 90 m.) Note who gets it right and who needs support.
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D. TEACHER REFLECTION

After the lesson, consider: Did learners hold the misconception that stopping distance is proportional to speed, and did the table change their minds? Which learners could not tell reaction distance from braking distance? Were learners confident in using $v = 0$ and $a = -5 \text{ m/s}^2$ in $v^2 = u^2 + 2as$, as practised in Lesson 3? Did the link between Kinangop and the slope make sense after Lesson 5? Was the road-crash discussion handled sensitively? Did every group choose a research topic (SGR train, pole vault or construction safety) with a clear linear-motion idea and a calculation? Which resources (devices or print) were available, and do some groups need printed support for Lesson 7? What would you change next time (for example, more time on the wet road or an extra practice problem)?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a matatu on Mombasa Road needs a longer total stopping distance as speed rises: on a dry road 20 m at 10 m/s (36 km/h), 60 m at 20 m/s (72 km/h) and 120 m at 30 m/s (108 km/h). The braking distances (10 m, 40 m, 90 m dry; 20 m, 80 m, 180 m wet) grew much faster than the speed.
What did I learn?	We learned that stopping distance = reaction distance + braking distance. Reaction distance is speed x reaction time (1.0 s), while braking distance comes from $v^2 = u^2 + 2as$ with $v = 0$, which gives $u^2/2 a $. So doubling the speed multiplies braking distance by 4 and tripling it multiplies it by 9, and a wet road ($a = 2.5 \text{ m/s}^2$) doubles the braking distance.
How does this explain the phenomenon?	We can explain that the runaway mkokoteni reaches 20 m/s (72 km/h) with no brakes and hits the lorry, while a matatu travelling at the same speed on a dry road needs 60 m to stop. Because braking distance depends on the square of speed, steep roads such as the Kinangop descent need warning signs and slow, early braking, and NTSA rules on speed and safe distance save lives. The same equations of linear motion that predict the cart's speed and distance also predict how far a vehicle needs to stop.

LESSON 7 (80 minutes (double lesson of 2 x 40 minutes)): Group Presentations and Building the Class Model: The Mkokoteni's Velocity-Time Graph

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners present their group research on real-life applications of linear motion (SGR train, pole vault, construction safety) and to represent the runaway mkokoteni's motion as a velocity-time graph (gradient = acceleration, area = displacement), linking all concepts from Lessons 1-6 into one class explanatory model.
Knowledge	Learners explain terms used in linear motion (distance, displacement, speed, velocity, acceleration). Learners calculate variables of motion using the equations of linear motion with constant acceleration: (1) $v = u + at$; (2) $s = ut + \frac{1}{2}at^2$; (3) $v^2 = u^2 + 2as$; (4) $s = \frac{1}{2}(u+v)t$. Learners investigate the effect of gravity on bodies under free fall. Learners appreciate the significance of linear motion and free fall in real life. For the cart: the v-t graph is a straight line through the origin with gradient 2 m/s^2 , the area under it from 0 to 10 s is $\frac{1}{2} \times 10 \times 20 = 100 \text{ m}$, and the s-t graph is a curve ($s = t^2$).
Skills	Plotting and reading a velocity-time graph from the mkokoteni table; calculating gradient and area under a graph; communicating research findings in a 3-minute presentation; constructing, using and revising a model (NGSS Developing and Using Models); constructing explanations from evidence (NGSS Constructing Explanations); using mathematics and computational thinking (NGSS); engaging in argument from evidence during cold-call follow-up questions.
Attitudes	Communication and collaboration: learners work with peers to discuss and present in plenary the application of linear motion in real life. Learning to learn: learners present findings from assignments in a group discussion. Peace: learners display tolerance and calmness when discussing the application of linear motion in real life. Unity: learners display cooperation and collaboration with others while building the class model. PCI Road safety: increased awareness on how to minimise fatal injuries resulting from road crashes on sloping roads using the concept of free fall of objects.
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	Lesson 7 pulls together the evidence from Lessons 1-6. Learners have the definitions (L1), the four equations (L2), multi-step problem solving and sign conventions (L3), g from measurement (L4), free fall and why the slope reduces the cart's acceleration to 0.2 of g (L5), and stopping distances and road safety (L6). Here they present their research and add one new idea: the velocity-time graph, where the gradient shows acceleration and the area shows displacement. They use it to build one model that answers the driving question: why the mkokoteni keeps getting faster and how we can predict its speed and distance before the crash. Lesson 8 then consolidates individually.
Safety Notes	No physical drops or moving carts are used in this lesson. Keep aisles clear when groups move to the model wall. Learners handle rulers and pencils only. Presenters should not stand on chairs to reach the model wall. If any group demonstrates a pole vault or similar action, it must be described or shown on paper only, not performed in the classroom.

B. LESSON OVERVIEW

Mzee Kamau's loaded mkokoteni (300 kg) rolled 100 m down the Limuru-Nairobi escarpment road and hit a parked lorry at 20 m/s (72 km/h). Achieng said, 'It started slowly then flew.' In this double lesson, learners first predict what a graph of the cart's velocity against time would look like and what its area might mean. They then observe three 3-minute group presentations (SGR train, pole vault, construction safety), with the teacher cold-calling 2 members per group for follow-up questions. Next they plot the full mkokoteni table, find the gradient (2 m/s^2) and the area (100 m at 10 s), and compare the s-t curve ($s = t^2$). Finally they review the Driving Question Board and build a class explanatory model (annotated diagram plus velocity-time graph) linking motion terms, equations, free fall, gravity and road safety back to the

cart.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Interpreting direction of motion from velocity-time graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Velocity-Time Graphs (Flexbook) Source: CK-12  Search ARES for similar readings	(10 min) Learners recall the video of the runaway mkokoteni and Achieng's words, 'It started slowly then flew.' Individually, then in pairs, they sketch on scrap paper what the graph of the cart's velocity (vertical axis) against time (horizontal axis) might look like for the 10 s down the escarpment, and write one sentence on what the steepness of the line would tell us. They also predict: 'If the cart travelled 100 m in 10 s and finished at 20 m/s, can we find the 100 m from the graph?' Learners write a prediction with a reason. They share in pairs and then hold up sketches.	Display the question: 'Draw velocity against time for Mzee Kamau's cart from the moment it is released until it hits the lorry.' Say: 'Remember, the cart has no engine and nobody is pushing. Think about what you know from Lessons 1 to 6. Do not use your notes. I want your honest prediction.' WAIT TIME 15 seconds in silence before pairs talk. Walk round and note sketches: straight line through origin, curve, flat line, line that starts high. Cold-call 3 learners with different sketches: 'Why did you draw it that way?' Do not correct. Record sketch types on the board with tally marks. Say: 'Hold on to your sketch. By the end of today we will test it against the evidence.'	Predict-Observe-Explain (Predict step) with a Gallery of Initial Ideas: making sketches public and tallied surfaces misconceptions (for example, a curve for velocity, or a flat line because the cart is not pushed) so that they can be tested against the table later. Learners' own ideas become the comparison point for later revision.	Observe and tally: (a) how many sketches show a rising line (consistent with Lesson 1's claim that velocity keeps increasing), (b) how many start at zero, (c) how many show a straight line versus a curve, (d) how many link the steepness to acceleration. Look for the misconception 'the graph shows the path of the road' (a hill shape). Record the tally on the board for comparison at the end of the lesson.
Observe Phase  VIDEO: Interpreting change in speed from velocity-time graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Velocity-Time Graphs (Flexbook) Source: CK-12  Search ARES	(25 min) Part A: Group presentations (3 groups x 3 minutes, plus 1-2 minutes of follow-up each). Groups present their Lesson 6 research on the SGR train, pole vault, and construction safety, each linking to the equations of linear motion, free fall or stopping distance. Listening learners complete a 3-column note sheet: 'Linear motion idea / Equation or term used / Link to the mkokoteni'. After each presentation, the 2 cold-called members answer follow-up questions. Part B: Data to graph. Using the full mkokoteni table ($t =$	Before presentations say: 'Presenters have exactly 3 minutes. Audience, your job is to find the link to Mzee Kamau's cart.' Keep time visibly. After each group, cold-call 2 members: 'Member A, which equation of linear motion did your group use and why?' 'Member B, how would your topic change if the road were wet?' WAIT TIME 10 seconds before each answer. If the first answer is incomplete, ask 'What would your partner add?' For the SGR train group ask: 'When the train brakes, is acceleration positive or negative? Why?' For	Evidence gathering through peer presentation and data representation (Analysing and Interpreting Data). Presentations broaden the evidence base from the cart to other real-world cases, while plotting the table converts numbers into a visual pattern. Using a graph to see that constant acceleration gives a straight line and that area equals distance builds the mathematical link between the table, the equations and the cart's motion.	Check note sheets: each learner should have at least 2 links to the cart per presentation. Observe whether the cold-called members state the correct equation and sign convention (a negative for braking; up positive for the vaulter). Check graphs: points plotted accurately to half a square, straight line through the origin, gradient calculated as 2 m/s^2 with correct units, area as 100 m with working shown. Listen for the common error of reading the gradient as 20 m/s (the final velocity) or as 10 m/s. Note learners who recognise that the area of 100 m matches the crash

for similar readings	0 to 10 s, $v = 0, 2, 4, 6, 8, 10, 12, 16, 20$ m/s at $t = 0, 1, 2, 3, 4, 5, 6, 8, 10$ s; $s = 0, 1, 4, 9, 16, 25, 36, 64, 100$ m), learners plot a v-t graph on graph paper (time on x-axis $1 \text{ cm} = 1 \text{ s}$; velocity on y-axis $1 \text{ cm} = 2 \text{ m/s}$) and draw the best-fit line. They find the gradient using the points (0, 0) and (10, 20): gradient = $20/10 = 2 \text{ m/s}^2$, and compare with $a = 2.0 \text{ m/s}^2$ from Lesson 1. Then they find the area of the triangle under the line from 0 to 10 s: $1/2 \times 10 \times 20 = 100 \text{ m}$, and compare with the table value $s = 100 \text{ m}$.	the pole vault group ask: 'At the top of the vault, what is the vertical velocity? What is a?' For the construction safety group ask: 'If a brick falls 20 m from a scaffold, how long does it take and how fast is it moving? (2 s, 20 m/s) How does that compare with the cart?' For Part B say: 'Plot every point. Use a ruler. Then I want you to tell me what the line is doing.' Circulate and ask: 'What is the gradient of your line? What are the units? What does it mean physically?' and 'How could you find the 100 m without using $s = t^2$?' WAIT TIME 15 seconds. Cold-call 3 learners for the gradient and 3 for the area.		distance.
Explain Phase  VIDEO: Interpreting direction of motion from velocity-time graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Average velocity and average speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(20 min) In groups of 4, learners use the graph to explain the cart's motion and answer: (1) Why is the line straight? (constant acceleration of 2.0 m/s^2, because gravity pulls it down the slope all the time: $a = g \times 0.2 = 10 \times 0.2$) (2) Why did Achieng say 'It started slowly then flew'? (velocity starts at 0 and rises 2 m/s every second; distance grows as t^2: 25 m by 5 s but 100 m by 10 s). (3) Use the graph to find the cart's velocity at 5 s (10 m/s) and the distance by 5 s as the area of a triangle ($1/2 \times 5 \times 10 = 25 \text{ m}$). (4) Why is the average speed 10 m/s, and where is that seen on the graph? (halfway up the line: $(0 + 20)/2$). (5) Sketch the s-t graph and explain why it is a curve ($s = t^2$). (6) Compare with the free fall of 20 m: a graph with a gradient of 10 m/s^2 would reach 20 m/s in 2 s, five times as steep. (7) Using the L6 table link, explain on a sentence strip why doubling speed means 4 times the braking distance, using the idea that the	Say: 'Use only evidence from your graph, the table and Lessons 1 to 6. I want a claim, evidence and reasoning.' Ask: 'The cart has a mass of 300 kg. If it carried 600 kg, would the gradient change? Why or why not?' WAIT TIME 15 seconds. (No: $a = g \times 0.2 = 2.0 \text{ m/s}^2$, mass does not change it.) Challenge: 'Someone says the cart is in free fall. Agree or disagree?' Cold-call 3 groups, one member each, using the stem 'I disagree because the acceleration is ____ not ____.' Address the misconception that a straight v-t line means constant velocity: 'Is this line flat? What does flat mean? What does sloping mean?' Prompt reasoning: 'How is the steepness of the free fall line different from the cart line, and what does that tell us about the slope?' Ask 2 members from every group to read their 3-sentence explanation aloud and have the class give a thumbs-up or thumbs-down signal for evidence use.	Claim-Evidence-Reasoning with Think-Pair-Share (Constructing Explanations). Learners use the graph's gradient and area as evidence for causal reasoning: gravity along the slope gives constant acceleration, so the line is straight. Contrasting the cart with free fall (same final speed of 20 m/s, 10 s versus 2 s) sharpens the idea that the slope reduces acceleration to 0.2 of g.	Look for explanations that include (a) constant acceleration of 2.0 m/s^2 caused by gravity along the slope, (b) gradient meaning acceleration, (c) area meaning displacement, and (d) why it feels sudden (s grows as t^2). Indicators of mastery: correctly states $v = 10 \text{ m/s}$ and $s = 25 \text{ m}$ at 5 s from the graph and checks them with $v = u + at$ and $s = \frac{1}{2}at^2$; states that mass does not change a; says the cart is not free fall. Misconceptions to note: using $a = 10$ for the cart; confusing the gradient with the area; saying heavier objects accelerate faster.

	area under a braking v-t graph is displacement. Each group writes a 3-sentence explanation answering 'Why does the cart keep getting faster, and how can we predict how fast and how far?'			
Driving Question Board (DQB) Creation  VIDEO: Interpreting change in speed from velocity-time graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Instantaneous velocity and instantaneous speed from graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(10 min) Learners return to the DQB started in Lesson 1. In groups, each takes 3-4 question cards and decides whether each is Answered, Partly answered, or Still open, giving the lesson evidence that supports the decision (for example, 'Why does the cart get faster if nobody pushes it?' answered by L1 and the gradient of 2 m/s^2 ; 'Can we predict how fast it will be?' answered by L2 equations and the table: $v = 20 \text{ m/s}$ at 10 s ; 'Why can't we stop in time on a slope?' answered by L6, with braking distance proportional to v^2). Learners move answered cards to the 'Answered' column with a sticky-note evidence statement and write any new questions (for example, 'What happens when friction is included?' or 'How do runaway-truck ramps on the Kinangop road work?') on new cards for Lesson 8 or further study.	Say: 'Read each question on the board aloud. Can we answer it using evidence from our lessons? Show me the evidence.' WAIT TIME 10 seconds before a group moves a card. Cold-call 4 learners from different groups: 'Which lesson gave you the evidence for that answer?' Push for specifics: 'Give me a number from the table.' Highlight questions that remain open: friction, air resistance and drag, and the real escarpment road. Respond to new questions: 'Great question, we ignored friction in our model. Add it to the Open column.' Link to KICD key inquiry question 2: 'How can the dangers of free fall in our day-to-day life be mitigated?' and point to the NTSA rules and the Kinangop warning signs from Lesson 6.	Question tracking on the Driving Question Board (Asking Questions): returning to the learners' own questions shows what the evidence has resolved and what has not, building ownership of the learning and showing that science progresses by refining questions.	Check that each card moved to 'Answered' has a correct evidence statement with a number or a named lesson. Listen for the three limits learners should recognise: the model ignores friction and air resistance, the cart is not free fall, and the numbers hold only for the steady 100 m straight section. Note the quality of new questions: those that refer to variables (slope angle, friction, mass, wet road) show deeper understanding.
Model Building Phase  VIDEO: Acceleration vs. time graphs <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	(15 min) The class builds one explanatory model on a large sheet on the wall: (1) a side-view diagram of the Limuru-Nairobi escarpment road with the mkokoteni at the top ($u = 0$), the lorry at the bottom 100 m along the road (20 m vertical drop), and the slope with $\sin(\text{angle}) = 0.2$ labelled; (2) arrows for the pull of gravity along the slope, labelled $a = g \times 0.2 = 2.0 \text{ m/s}^2$; (3) time stamps along the slope with v and	Say: 'Our model must answer the driving question: why does the cart keep getting faster, and how can we predict how fast and how far? Every part of the model must connect to another part with an arrow and a reason.' Assign components to groups: Group 1 diagram and slope; Group 2 forces and acceleration; Group 3 v-t graph; Group 4 s-t curve and table; Group 5 free fall comparison; Group 6 road safety. WAIT TIME	Collaborative Model Building and Revision (Developing and Using Models): the model integrates the motion terms, the equations, graphs, gravity, free fall and safety into one coherent account. Revising the initial sketch makes learning visible and shows that models are improved by evidence.	Use a checklist: model includes a labelled slope with $\sin(\text{angle}) = 0.2$, $a = 2.0 \text{ m/s}^2$ (not 10), table values ($v = 20 \text{ m/s}$, $s = 100 \text{ m}$ at 10 s), the straight v-t line with gradient 2 m/s^2 and area 100 m , the curved s-t graph, the free fall comparison (2 s versus 10 s , both 20 m/s), and the road safety link. Observe whether learners can trace a causal chain: gravity along the slope, then constant acceleration, then velocity rising 2

 HTML: Calculating Acceleration from Velocity and Time (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	s from the table ($t = 0, 5, 10$ s: 0, 10, 20 m/s; 0, 25, 100 m); (4) the velocity-time graph with gradient 2 m/s^2 and the shaded area of 100 m, and a small s-t curve; (5) a call-out box for free fall: a vertical drop of 20 m takes 2 s, $a = 10 \text{ m/s}^2$, final $v = 20 \text{ m/s}$; (6) a call-out box for road safety: stopping distance = reaction distance + braking distance, doubling speed multiplies braking distance by 4, with NTSA rules and warning signs like those on the Kinangop descent. Each group adds one labelled component, and the class agrees on the wording. Learners then compare with their initial predictions from the Predict Phase and revise their own sketches in a different colour.	15 seconds before each group adds its piece, and ask the class, 'Does this match our evidence? Raise a hand if you disagree.' Cold-call 3 learners to trace a path through the model: 'Start at gravity and end at the crash. Tell me the story using the model.' Ask: 'What does the model not yet explain?' (friction, air resistance, the real road). Close by returning to the prediction tally: 'How many of you changed your sketch? What evidence changed your mind?' Preview Lesson 8: 'Next lesson you will explain the cart on your own, so keep this model in mind.'		m/s every second, then distance growing as t^2, then a crash at 72 km/h. Check that revised sketches show clear changes with a written reason.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Could every learner explain, in their own words, why the v-t line for the cart is straight and what its gradient and area mean? (2) Which group presentation best connected its topic to the cart, and which needed more help to link the equations of linear motion to their example? (3) Did any learners still use $a = 10 \text{ m/s}^2$ for the cart or call it free fall, and which question exposed it? (4) Were the 3-minute limits and cold-calls managed fairly, so quieter learners spoke? (5) Did the model-building phase allow every learner to contribute? (6) Which DQB questions remain open, and what should Lesson 8 emphasise for the individual written explanation? Note learners who need support with graph plotting, reading the area as a triangle, or unit conversion (km/h divided by 3.6 gives m/s) before Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe about the mkokoteni's motion from the table and graph? (For example: velocity rises by 2 m/s every second, giving a straight line through the origin on the v-t graph; distance grows as t^2 , with $s = 25 \text{ m}$ at 5 s but 100 m at 10 s; the cart hits the lorry at 20 m/s, or 72 km/h.) What did you observe in the group presentations about linear motion in real life?
What did I learn?	What did we learn today? (For example: the gradient of a velocity-time graph is acceleration (2 m/s^2); the area under it is displacement ($\frac{1}{2} \times 10 \times 20 = 100 \text{ m}$); a constant acceleration gives a straight line; the s-t graph is a curve; free fall is the same physics with $a = g = 10 \text{ m/s}^2$; stopping distance rises sharply with speed.)
How does this explain the phenomenon?	How do we now explain the driving question? (For example: gravity pulls the mkokoteni down the slope all the time, giving a constant acceleration of $a = g \times 0.2 = 2.0 \text{ m/s}^2$, so its velocity keeps rising from 0 to 20 m/s in 10 s; it starts slowly because it starts from rest and feels sudden because distance grows as t^2 ; the equations of linear motion and the v-t graph let us predict speed and distance at any moment, and they show why higher speeds need much longer stopping distances.)

LESSON 8 (80 minutes (2 x 40-minute lessons)): Consolidation, DQB Review and Final Explanation: Why the Mkokoteni Keeps Getting Faster and How We Predict It

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate learning from Lessons 1-7 through individual problem solving and a written Final Explanation of the mkokoteni phenomenon, to close the Driving Question Board, and to reflect on how thinking has changed since Lesson 1. No new content is taught.
Knowledge	By the end of the lesson, the learner should be able to: a) explain terms used in linear motion (distance, displacement, speed, velocity and acceleration) b) calculate variables of motion using equations of linear motion with constant acceleration c) investigate the effect of gravity on bodies under free fall d) appreciate the significance of linear motion and free fall in real life. Content used: the mkokoteni table ($t=10$ s, $v=20$ m/s= 72 km/h, $s=100$ m, $a=2.0$ m/s ²), the free fall comparison (20 m drop: 2 s and 20 m/s in free fall versus 10 s and 20 m/s for the cart), the mango falling from a 5 m branch ($t=1$ s, $v=10$ m/s), and a sprinter problem.
Skills	Selecting an equation by listing knowns and unknown; applying the sign convention; using the equations with $u=0$; interpreting the velocity-time graph (gradient = acceleration, area = displacement); constructing a written scientific explanation (claim, evidence, reasoning); comparing an early model with a final model; reflecting in a journal. NGSS practices: Constructing Explanations, Developing and Using Models, Analyzing and Interpreting Data, Using Mathematics and Computational Thinking, Engaging in Argument from Evidence.
Attitudes	Core competencies: Critical thinking and problem solving (solving the structured problem set); Learning to learn (reflecting on how own thinking changed); Communication and collaboration (the whole-class DQB review). Values: Peace (tolerance and calmness when discussing different answers); Unity (cooperation during the DQB review). PCIs: Road safety (minimising fatal injuries from crashes on sloping roads); Life Skills Education (self-awareness through honest reflection on own learning).
Key Inquiry Question	1. How is free fall motion important in real life? 2. How can the dangers of free fall in our day-to-day life be mitigated?
Purpose in Storyline	This is the final lesson of the storyline. Learners pull together everything from Lessons 1-7 to answer the Driving Question: why the runaway mkokoteni on the Limuru escarpment keeps getting faster, and how mathematics predicts exactly how fast and how far it travels before it crashes. It builds on the Lesson 7 class model (annotated diagram plus velocity-time graph) and the DQB, and it shows how far thinking has moved from the Lesson 1 predictions.
Safety Notes	This is a paper-and-discussion lesson, so there are no hazardous activities. Keep the classroom clear of bags while learners move to the DQB. Do not allow learners to test falling objects from desks or windows. Remind learners that the road-safety discussion is about Mzee Kamau's cart, and be sensitive: no one was hurt, but the cart was destroyed.

B. LESSON OVERVIEW

Learners begin by revisiting their Lesson 1 predictions about Mzee Kamau's runaway mkokoteni and make a brief new prediction about a new situation, a sprinter starting from rest. They then work individually through a structured problem set set in the anchoring phenomenon and its supporting phenomena: the full mkokoteni table ($a=2.0$ m/s², $v=20$ m/s and $s=100$ m at $t=10$ s), the free fall comparison (a 20 m vertical drop takes 2 s and gives 20 m/s, but the cart needs 10 s to reach the same 20 m/s because the slope provides only 0.2 of g), the mango falling from a 5 m branch, and a Rift Valley sprinter built like the Lesson 3 runner problem. In the Explain phase learners write an individual Final Explanation (claim, evidence, reasoning) answering the Driving Question. The class then completes the DQB: each remaining question is discussed and answered collectively, with the evidence cited. Finally, learners compare a Lesson 1 model with a final model and write a journal reflection. Achieng's words, 'it started slowly then flew', are used as a thread to check whether learners can now explain what she saw.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Choosing kinematic equations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Setting up problems with constant acceleration Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners open their Lesson 1 journals and read their original prediction about why Mzee Kamau's 300 kg mkokoteni kept getting faster. In pairs (2 minutes) they underline any idea they now think was incomplete. Then each learner writes a quick prediction (individually, 3 minutes) for a NEW case: a Rift Valley 400 m runner starts from rest and accelerates at 4 m/s^2 for 2.5 s. They predict whether she will be running faster than 5 m/s, 10 m/s or 20 m/s, and roughly how far she will have gone. They also predict: if the cart were dropped straight down a 20 m vertical drop instead of rolling down the slope, would it reach the bottom faster or slower than 10 s, and would its final speed be greater, smaller or the same as 20 m/s?	Say: 'Open your journals to Lesson 1. Read what you wrote about why the cart kept getting faster. Does it still make sense to you?' WAIT TIME 10-15 seconds in silence. Cold-call 3 learners: 'What did you write in Lesson 1?' Do not correct them yet. Then say: 'Today you are the experts. Before we calculate, predict: the runner starts from rest and accelerates at 4 m/s^2 for 2.5 s. Is her speed after 2.5 s closer to 5, 10 or 20 m/s?' WAIT TIME 10-15 seconds. Ask all learners to show 1, 2 or 3 fingers for their choice, and record the tally on the board. Then ask: 'If the cart fell straight down a 20 m drop, faster or slower than 10 s?' Cold-call 2 learners to give reasons. Record predictions and say: 'We will check these against the evidence by the end of the lesson.'	Prediction with Revisiting of Prior Ideas: learners confront their own earlier thinking and commit to new predictions before calculating. Committing to a prediction makes a later correct calculation more memorable, and it exposes any remaining misconceptions (for example, that heavier carts accelerate faster or that speed must come from a continuing push).	Observe whether learners (a) can locate and read their Lesson 1 prediction, (b) justify their runner prediction using the idea of acceleration as change in velocity per second rather than a vague 'she is fast', and (c) mention gravity or the slope in the cart prediction. Tally the finger votes and note the 3 most common misconceptions on the side board: 'a force is needed to keep moving', 'mass changes the acceleration', 'a steeper path gives a different final speed'. Use the tally to decide which problems to prioritise during circulation.
Observe Phase  VIDEO: Deriving displacement as a function of time, acceleration, and initial velocity Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Displacement During Constant Acceleration (Flexbook) Source: CK-12  Search ARES for similar	Individually, learners complete the structured 'Mkokoteni Problem Set' (about 25 minutes) in their exercise books, using $g = 10 \text{ m/s}^2$ and the sign convention. They must list the knowns (u , v , a , s , t), name the unknown, choose the equation, and show units. Part A (the cart, $a=2.0 \text{ m/s}^2$, $u=0$): (1) Use $v = u + at$ to find the cart's velocity at $t = 2 \text{ s}$ and $t = 5 \text{ s}$ (4 m/s , 10 m/s). (2) Use $s = ut + \frac{1}{2}at^2$ to find its distance at $t = 5 \text{ s}$ and $t = 10 \text{ s}$ (25 m , 100 m). (3) Use $v^2 = u^2 + 2as$ to check the speed after 100 m ($v^2 = 400$, so $v = 20 \text{ m/s}$) and convert to km/h (72 km/h). (4) Use $s = \frac{1}{2}(u+v)t$ to confirm the average speed over 10 s (10 m/s). Part B (supporting phenomena): (5) The	Say: 'This is your chance to show what YOU can do, not what your group can do. Silent, independent work for the first 20 minutes.' Circulate with a clipboard and check the first line of each learner's working: 'Show me your list of knowns. Which one is missing?' Do not give answers; use prompts such as 'What is u for a cart released from rest?' and 'Why is $a = 2$ and not 10 for the cart?' For learners making the error of using $a = 10$ for the cart, ask: 'Is the cart in free fall? What fraction of g does the slope give?' WAIT TIME 10-15 seconds after every prompt before saying anything else. After 20 minutes allow learners to compare final	Individual Problem Solving with Structured Data (Using Mathematics and Computational Thinking): working through the same cart data in several equation forms shows that one idea, constant acceleration, produces consistent predictions (the answers $v = 20 \text{ m/s}$ and $s = 100 \text{ m}$ agree whichever equation is used). Comparing the 2 s free fall with the 10 s cart descent makes the role of the slope visible.	Check that each learner: (a) lists knowns and the unknown before calculating; (b) uses $a = 2.0 \text{ m/s}^2$ for the cart and $a = 10 \text{ m/s}^2$ only for the mango and the vertical drop; (c) reports units and the correct sign; (d) gets $v = 20 \text{ m/s}$ and $s = 100 \text{ m}$ for the cart at $t = 10 \text{ s}$; (e) gets 10 m/s and 12.5 m for the runner; (f) draws a straight line through the origin for the velocity-time graph with gradient 2 m/s^2 . Tick on a class checklist: 'Selects correct equation', 'Correct a ', 'Correct units', 'Graph correct'. Learners with fewer than 5 of 8 correct are flagged for a short follow-up conversation.

readings	mango falls from a 5 m branch of a mango tree in a school compound with $u=0$ and $a=g$: find t and v (1 s, 10 m/s). (6) Free fall comparison: find the time and speed for a 20 m vertical drop (2 s, 20 m/s) and explain why the cart needs 10 s to reach the same 20 m/s. (7) The Rift Valley 400 m runner starts from rest and accelerates at 4 m/s^2 for 2.5 s: find v and s (10 m/s, 12.5 m). Part C (graph): (8) Sketch the cart's velocity-time graph for 0-10 s, mark the gradient (2 m/s^2) and calculate the area ($\frac{1}{2} \times 10 \times 20 = 100 \text{ m}$). Learners who finish early attempt a challenge: at what time is the cart exactly half-way (50 m) down the slope? ($t = \sqrt{50}$) which is about 7.1 s, and v is about 14 m/s). Learners who need support use the quick-reference card of the four equations ($v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$; $s = \frac{1}{2}(u+v)t$) and the 'list knowns' organiser from Lesson 2.	answers in pairs for 5 minutes only. Stop at 25 minutes and cold-call 4 different learners to state one answer each (cart at 5 s, mango, runner, graph area) and ask the class to give a thumbs up/down for agreement. Do not collect marks yet; the answers become evidence for the Explain phase.		
Explain Phase  VIDEO: Average velocity for constant acceleration Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Acceleration Due to Gravity (Flexbook) Source: CK-12  Search ARES for similar readings	Learners write the Final Explanation individually (about 15 minutes) in answer to the Driving Question: 'Why does a runaway mkokoteni on the Limuru escarpment keep getting faster, and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes?' They use a Claim-Evidence-Reasoning frame: CLAIM: a one-sentence answer. EVIDENCE: at least three numbers from the mkokoteni table (for example 4 m/s at 2 s, 10 m/s at 5 s and 25 m, 20 m/s (72 km/h) and 100 m at 10 s), plus the free fall comparison. REASONING: link the evidence to the science: gravity pulls the cart down the	Say: 'Now write the explanation you would give Mzee Kamau if he asked why his cart flew away from him.' Display the CER frame and the checklist on the board (gravity; constant acceleration of 2.0 m/s^2; numbers from the table; $s = t^2$; equations; free fall comparison). Remind learners: 'Use the words and numbers from our class model, not just the formula.' Circulate and read over shoulders; ask 'Where in your writing do you explain WHY it keeps getting faster when nobody is pushing?' WAIT TIME 10-15 seconds, then let the learner revise. Ask 2 learners with different strengths to read their reasoning aloud; cold-call 3 more to name one piece of	Claim-Evidence-Reasoning (Constructing Explanations): learners must connect numerical evidence to causal mechanisms (gravity gives constant acceleration), which turns calculation into explanation. Peer checking with a shared checklist supports Engaging in Argument from Evidence and lets learners calibrate their own writing against the criteria.	Score each Final Explanation 0-3 on four criteria: (1) Cause: identifies gravity acting down the slope all the time (no engine or push needed); (2) Constant acceleration: states 2.0 m/s^2 and that velocity rises by 2 m/s every second; (3) Evidence: uses at least 3 correct values from the table ($v=20 \text{ m/s}$ and $s=100 \text{ m}$ at 10 s) and the free fall comparison; (4) Prediction: states that the equations of linear motion let us calculate speed and distance at any time. Exceeds: all four at level 3 and explains both 'starts slowly' (v begins at 0) and 'feels sudden' ($s = t^2$). Meets: all four at level 2 or higher. Approaches: gravity named but acceleration or numbers

	slope all the time; this gives a constant acceleration of 2.0 m/s^2 ($g \times 0.2$) so velocity rises by 2 m/s every second; it starts slowly because velocity begins at zero; it feels sudden because $s = t^2$, so 25 m by 5 s but 100 m by 10 s; the equations of linear motion predict speed and distance at any moment; free fall is the same physics with $a = g = 10 \text{ m/s}^2$ on a vertical path; the cart takes 10 s for the same 20 m/s that free fall reaches in 2 s because the slope provides only 0.2 of g. They must also explain Achieng's statement, 'it started slowly then flew'. Learners include a labelled sketch with the velocity-time graph. In the last 3 minutes they swap with a partner, who uses a checklist to say whether the explanation includes gravity, constant acceleration, the equations, and the free fall comparison.	evidence they used. Explicitly avoid giving a model answer until the partner check is done. During the partner swap say: 'Tick, do not rewrite. Give your partner one thing that is strong and one thing missing.' Collect the explanations for scoring with the rubric.		incomplete. Below: describes the motion without a cause. Watch for the misconception that the cart's mass makes it accelerate more (acceleration is 2.0 m/s^2 regardless of mass) and for any use of $a = 10$ for the cart.
Driving Question Board (DQB) Creation  VIDEO: Choosing kinematic equations Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Acceleration Due to Gravity Practice Source: CK-12  Search ARES for similar readings	The class conducts the final DQB review (about 15 minutes). The board has the questions from Lesson 1 and those added during Lessons 2-7 (for example: Why does it start slowly? Does the heavy maize make it faster? Would a steeper road be worse? How is free fall different from the cart? How can the dangers be reduced?). Learners, in rows, each take one question from the board. Individually they write a 1-2 sentence answer on a sticky note with the evidence (lesson and number) and place it next to the question. The class then discusses the questions that remain unanswered or disputed, and answers them collectively. Learners link the final two KICD inquiry questions: (1) How is free	Say: 'Stand up and look at our board. Pick one question you can now answer with evidence. Write the answer and the number that supports it.' WAIT TIME 10-15 seconds before learners move. For each question that stays blank, ask: 'Who can answer this using something we measured or calculated?' Cold-call 5 learners to read and defend a note, and use the prompt 'What evidence tells us that?' For disputed answers, invite a second learner to challenge respectfully (Peace and Unity): 'Do you agree, and what number supports you?' Mark each question as 'Answered', 'Partly answered' or 'New question' with different coloured markers. Say: 'Which question did we not fully answer? Why? What would we need to	Driving Question Board Review (Engaging in Argument from Evidence): returning to the learners' own questions shows how far the class's understanding has grown, and requiring evidence for each answer prevents unsupported claims. Recording questions that remain open models how scientific inquiry continues.	Look for: (a) every learner placing an answered note with the supporting evidence, (b) answers that cite specific numbers (for example '20 m/s at 10 s', '$a = g \times 0.2 = 2.0 \text{ m/s}^2$'), (c) learners who respectfully challenge or add to a peer's answer, (d) reasonable answers to the two KICD inquiry questions that connect free fall and speed to road safety and NTSA rules. Count the questions answered and partly answered and record the percentage for the class record. Note which learners still ask 'why does it keep speeding up?', as this signals a need for one more conversation.

	fall motion important in real life? (for example, estimating drop heights from impact speed such as the 45 m drop from a 30 m/s impact, the mango tree, pole vault and construction safety); (2) How can the dangers of free fall and fast descent be mitigated? (road signs on the Kinangop descent, NTSA rules, lower speeds because braking distance grows with the square of the speed, safe handling of mkokotenis on slopes, helmets and barriers). Any question that cannot be answered by the lesson content is moved to a 'Future Questions' section for further investigation (for example, the effect of friction and air resistance on the cart).	investigate?' to model that science leaves open questions. End by reading the final status aloud: for example 'We have answered all questions about speed and distance; friction and air resistance remain for future study.'		
Model Building Phase  VIDEO: Deriving displacement as a function of time, acceleration, and initial velocity Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Setting up problems with constant acceleration Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners complete the model comparison and journal reflection (about 15 minutes). Step 1 (5 minutes): in their journals they draw two diagrams side by side. LEFT: their Lesson 1 model/prediction of the cart (copied from the journal). RIGHT: their final model, an annotated diagram of the Limuru-Nairobi slope showing the 100 m of road, the vertical drop of 20 m, the force of gravity pulling the cart down the slope, the constant acceleration of 2.0 m/s^2, velocity labels at $t = 0, 2, 5$ and 10 s ($0, 4, 10$ and 20 m/s), the distance labels ($0, 4, 25$ and 100 m), the lorry at the bottom, and the velocity-time graph (straight line through the origin, gradient 2 m/s^2, area 100 m). A small inset shows the free fall comparison (20 m drop: 2 s, 20 m/s). Step 2 (10 minutes): learners write a journal reflection using these starters: 'In Lesson 1 I thought... Now I know... The	Say: 'Compare your first model with your final model. What has changed? What stayed the same?' WAIT TIME 10-15 seconds before learners begin writing. Circulate and ask: 'Where in your final model can I see why the speed keeps increasing?' and 'Show me where the model predicts 20 m/s.' Insist on labels with units. Select 3 learners (cold-call) to share one sentence from their reflection beginning 'In Lesson 1 I thought... Now I know...'. Return to the board tally from the Predict phase and ask: 'Which predictions were right? Which changed?' Reveal the answers (runner 10 m/s and 12.5 m; the 20 m vertical drop is faster at 2 s and gives the same 20 m/s) and ask learners to say why. Close with: 'Gravity does not need an engine or a push. A constant acceleration of 2.0 m/s^2 turns 0 m/s into 72 km/h in just 10 s. That is why we respect steep roads and why NTSA signs matter.' Collect	Model Comparison and Reflective Journaling (Developing and Using Models; Learning to Learn): placing the early model next to the final model makes conceptual change visible, and writing about the evidence that changed their minds helps learners to consolidate the causal story (gravity, constant acceleration, equations, free fall) and see themselves as people who can predict real events.	Final model checklist: (1) gravity or the slope shown as the cause; (2) acceleration 2.0 m/s^2 stated with units; (3) at least three correct velocity or distance values from the table; (4) velocity-time graph with gradient 2 m/s^2 and area 100 m; (5) free fall comparison shown; (6) a clear difference from the Lesson 1 model. Journal indicators: names at least one specific idea that changed, cites evidence, and links to road safety. Exceeds: all six features plus a genuine new question. Meets: at least five features. Approaches: three or four. Below: fewer than three. Record the result in the learner's profile as evidence for Sub-Strand 3.2.

	evidence that changed my mind was... The role of gravity is... The role of constant acceleration is... The equations of linear motion help us because... If I could speak to Mzee Kamau, I would tell him... Something I still wonder about is... Learners also revisit the opening predictions (runner 10 m/s; cart 20 m drop faster than 10 s and the same 20 m/s) and mark them correct or incorrect with an explanation.	journals for the portfolio.		
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D. TEACHER REFLECTION

After the lesson, consider: (1) How many learners could explain BOTH 'it started slowly' (velocity begins at zero) AND 'it feels sudden' (distance grows as t^2) without prompting? (2) Did any learner still use $a = 10 \text{ m/s}^2$ for the cart or suggest that heavier loads accelerate faster? If so, which activity from Lessons 4-5 should be revisited? (3) Compare the Lesson 1 and final models for 5 learners: what changed most (cause, numbers, graph)? (4) Which DQB questions stayed open, and should they become a future project (for example friction on the escarpment road, SGR braking, pole vault)? (5) Were the conditions for the individual problem set fair, with enough time and a quick-reference card for learners who needed support? (6) How did Peace and Unity show during the DQB review? Record the learners who need extra support on equation selection and plan a short clinic before the end-of-strand assessment.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the mkokoteni released from rest on the Limuru-Nairobi escarpment slope speeds up steadily: 0 m/s at 0 s, 4 m/s at 2 s, 10 m/s at 5 s and 20 m/s (72 km/h) at 10 s after 100 m. We observed that a mango dropped from a 5 m branch reaches 10 m/s after 1 s, and that objects dropped from 2.0 m took about 0.64 s regardless of what they were.
What did I learn?	We learned that velocity increases by 2 m/s every second because the cart has a constant acceleration of 2.0 m/s^2 ($g \times 0.2$), caused by gravity pulling it down the slope all the time. We learned the four equations of linear motion, how to choose the right one by listing knowns and the unknown, and that on a velocity-time graph the gradient is the acceleration and the area is the displacement. We learned that free fall is the same motion with $a = g = 10 \text{ m/s}^2$ on a vertical path.
How does this explain the phenomenon?	The mkokoteni keeps getting faster even with no one pushing because gravity acts on it down the slope all the time, giving a constant acceleration of 2.0 m/s^2 . It starts slowly because its velocity begins at 0 m/s, and the crash feels sudden because the distance grows as t^2 (25 m by 5 s but 100 m by 10 s). Using $v = u + at$, $s = ut + \frac{1}{2}at^2$ and $v^2 = u^2 + 2as$ we can predict that it reaches 20 m/s (72 km/h) after 100 m and 10 s. In a 20 m free fall the same 20 m/s is reached in just 2 s, because the slope gives the cart only 0.2 of g . This is why high speeds need much longer stopping distances and why road-safety rules on steep roads matter.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.